

VayuNetra — User Guide & Project Reference

The operations layer for urban air quality — trace who is polluting, predict what comes next, act with cited evidence, protect citizens in their own script. Ten Indian cities, ₹0 infrastructure.

Version of 18 August 2026 · Live app <https://vayunetra-aqi.vercel.app> · Source github.com/omkarr88/VayuNetra · Team DaGoats.

Every screenshot in this guide was captured by an automated walkthrough of the live console on 18 August 2026 (`web/scripts/qa/full-walkthrough.mjs`), so what you see here is what the application does — the same walkthrough checks every control and every download and passed with zero errors.

1. What VayuNetra is, in one minute

Indian cities already **measure** their air (900+ CAAQMS stations) and several **forecast** it. Nothing **acts** on it: a reading turns red and no system tells a city officer *who* is polluting this square kilometre right now, *what* the air will do tomorrow, or *where* to send inspectors first.

VayuNetra is that missing operations layer. Every ~1 km² cell in a city (an H3 resolution-8 hexagon) carries four things:

Verb	What the system does	Where you see it
Trace	Assigns PM2.5 blame per cell to traffic / construction dust / industrial / biomass burning / transported / other — a gradient-boosted model with SHAP explanations blended with chemical-signature priors, which abstains where it lacks out-of-sample skill	the map, the Cell Story
Predict	PM2.5 at +24 / 48 / 72 h with an 80 % conformal band and a calibrated probability of crossing Very Poor (>120 µg/m ³) or Severe (>250)	Cell Story tiles, Forecast section
Act	Ranks emission sources by contribution × people exposed × actionability × confidence; each with the real Sentinel-2 image of the site, the regulation retrieved verbatim, a draft notice PDF; the officer approves → dispatches → closes the case with a field finding on an immutable audit trail; every dispatch is measured before/after	Enforcement section
Protect	Templated (LLM-free) health advisories in eight scripts — Hindi, Kannada, Marathi, Tamil, Telugu, Bengali, Gujarati, English — over the app, a live Telegram bot, an IVR line and public displays	Advisories section

Six AI agents on one LangGraph do this in seconds of compute; the human steps (approval, dispatch, closure) are timestamped separately. It runs live in **Delhi, Bengaluru, Mumbai, Hyderabad, Chennai, Kolkata, Pune, Ahmedabad, Jaipur, Lucknow** on free tiers (Vercel + Render + Supabase + GitHub Actions). Adding a city is one YAML file.

VayuNetra

How it worksArchitecturePlatformValidation

Open console

ET AI HACKATHON 2026 · PROBLEM STATEMENT 5

The operations layer for urban air quality.

India already measures its air — 900+ CAAQMS stations — yet a 2024 CAG audit found only 31% of monitored cities have any actionable response protocol. VayuNetra is that missing layer: it traces PM2.5 to its sources square-kilometre by square-kilometre, forecasts 72 hours ahead with calibrated uncertainty, and turns both into cited enforcement notices and citizen alerts in eight languages. Live today across 10 Indian cities — from Delhi to Lucknow — built entirely on free public infrastructure.

Open the console →

How it works

live · Delhi AQI

79 Satisfactory

last pipeline run 1.4s end-to-end

vayunetra-aqi.vercel.app/console

VayuNetra

Delhi

Enforcement

Forecast

Advisories

UNDERSTAND

Cities

79 Satisfactory

Worstoff: PM2.5 +3 µg/m³

data 2h ago · LIVE

CDAP Stage 1: from forecast

Queue traffic: 18 slots

LAST PIPELINE RUN

0.7s

signal-action: 5 min off

signal - 10 min - 1 hour to 72h up

Layers

ACT Enforcement

Where to send an Inspector today, with the evidence to back it — from the worst places on the map to a signed notice and a tracked outcome.

1 Worst places now

2 Dispatch by ward

3 Evidence & notice

4 Track outcomes

Enforcement Worklist

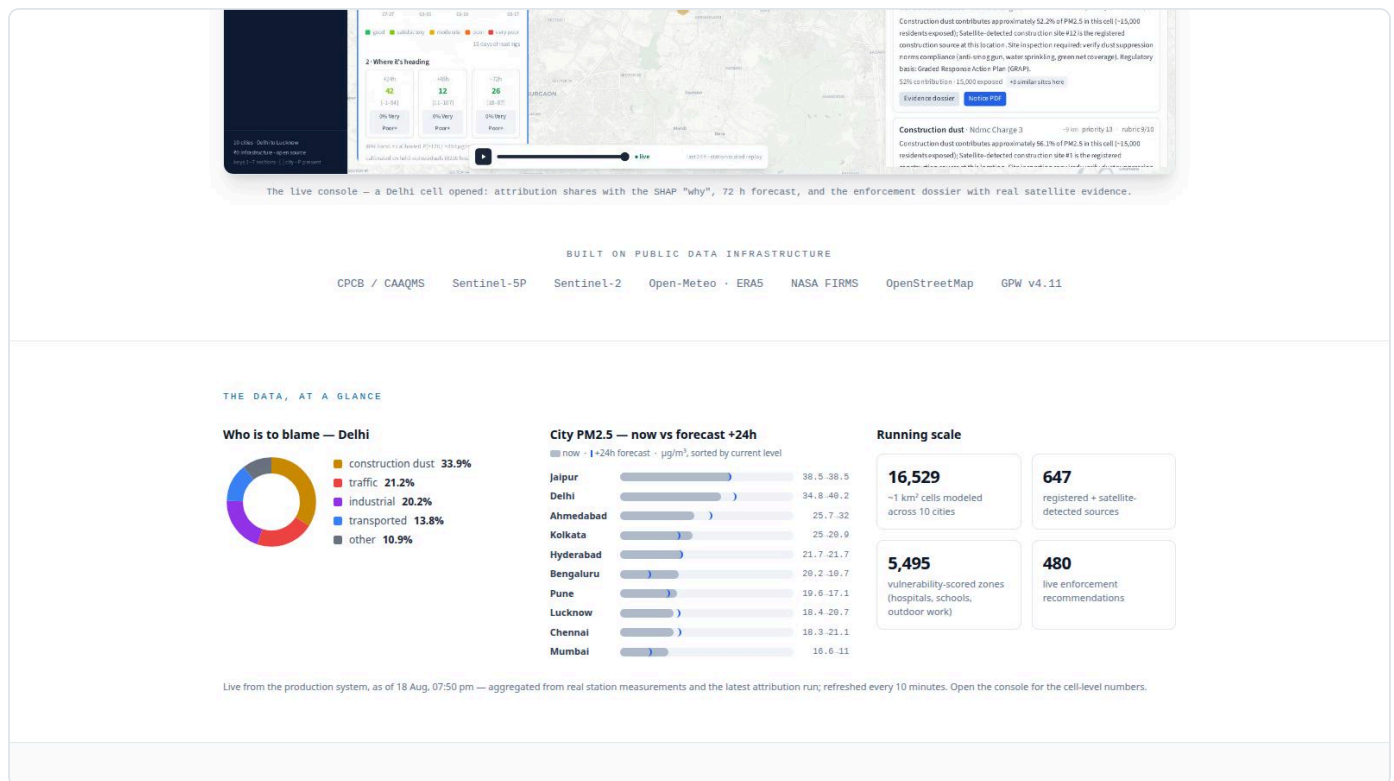
Inspected to selected call End

2. Getting in

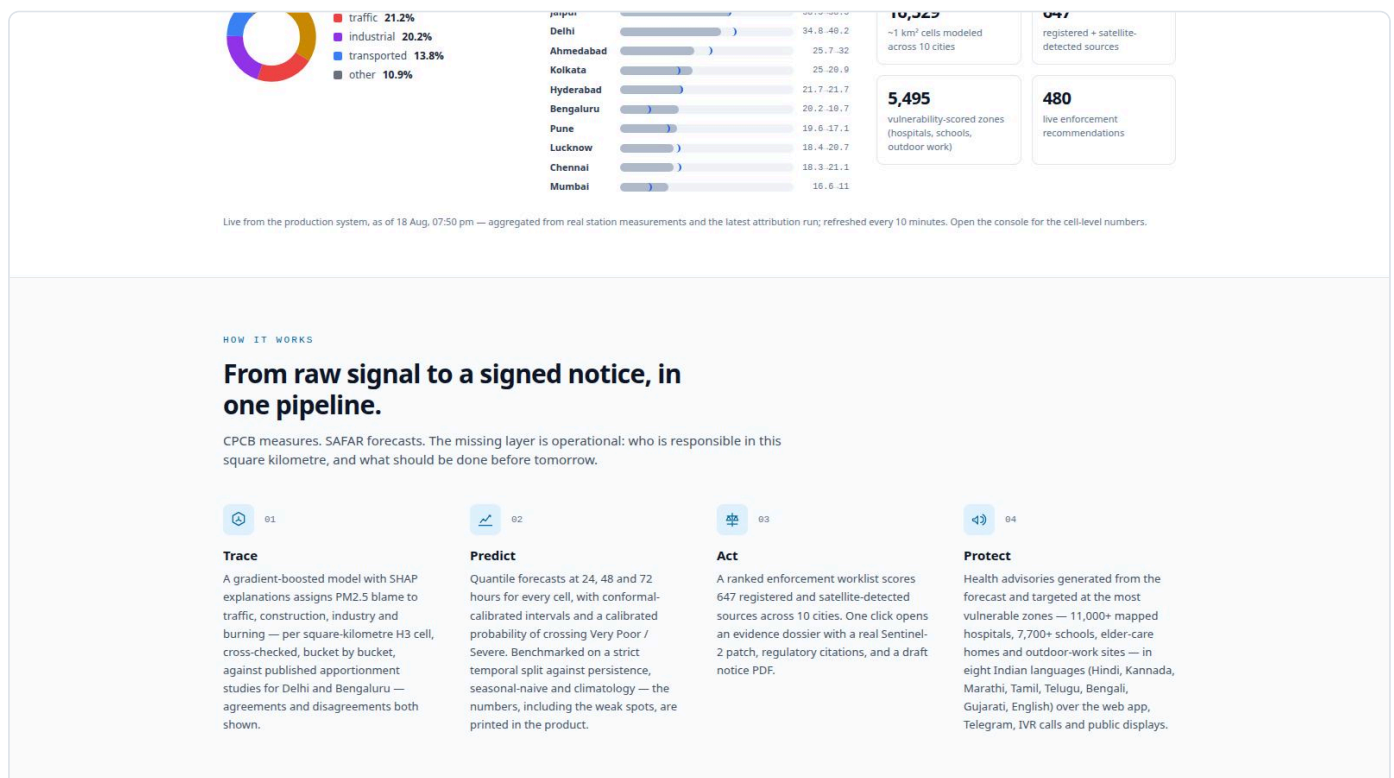
Public app	https://vayunetra-aqi.vercel.app → Open console . No login: reads are public under Postgres row-level security; every write goes through the server with the service role.
API	https://vayunetra-c8i8.onrender.com — GET <code>/health</code> shows <code>DEMO_MODE</code> ; every endpoint returns <code>{success, data, error, meta}</code> . Full list: <code>docs/API_CONTRACT.md</code> .
Run locally	<code>make dev</code> — starts the API on <code>:8000</code> (<code>DEMO_MODE</code> from <code>.env</code>) and the web app on <code>:5173</code> . <code>make prewarm</code> wakes and verifies the live stack (GO / NO-GO). <code>make test</code> runs the 204 backend tests.
Demo mode	<code>DEMO_MODE=true</code> serves 19 bundled fixtures with an identical UX (labelled in <code>/health</code>). The web app also falls back to bundled fixtures if the API is unreachable and shows an amber <i>"backend waking up — showing bundled demo snapshot"</i> notice (Render free tier sleeps; retry or wait — it clears on the first successful call).
Keyboard	In the console: 1–7 switch sections · [] cycle cities · P presentation mode (larger type for a projector) · ? replays the tour · Esc closes the Cell Story.
Deep links	Every console state is a URL: <code>/console?city=mumbai&section=forecast&cell=88608b56cbfffff&mode=pm25&layers=sources,plumes,wards,freight,fires</code> .

3. The landing page (/)

The public front door — for citizens, judges and officers who have not opened the console yet. It reads live data where it can.



- **Hero** — the one-sentence proposition, **Open the console**, and the console screenshot.
- **The data, at a glance** — read live from `GET /landing/snapshot` (10-minute cache, stamped *as of HH:MM*): the Delhi source-mix donut (mean of the latest attribution run), every city's PM2.5 now vs +24 h forecast (sorted worst first), and the running scale — ~1 km² cells modelled, registered + satellite-detected sources, vulnerability-scored zones, live enforcement recommendations. A bundled snapshot is shown only until the API answers.

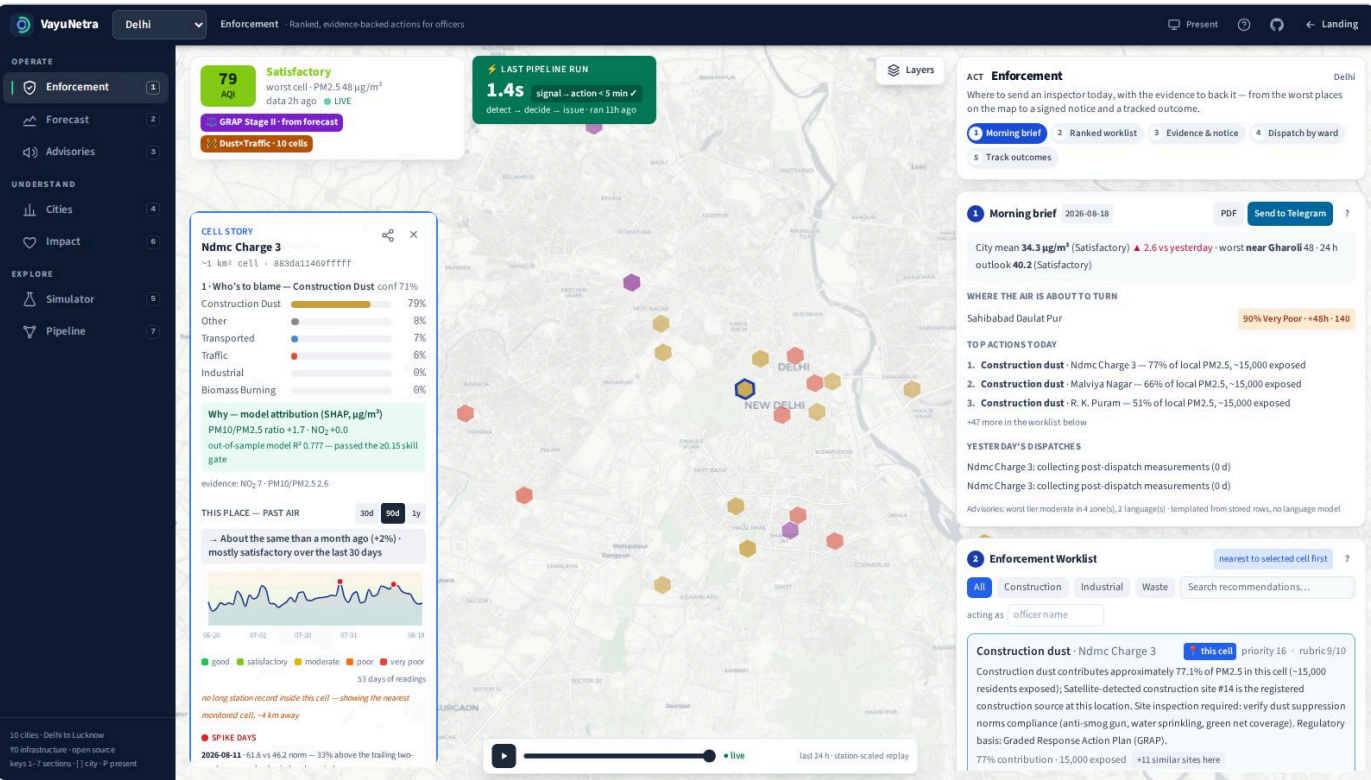


- **How it works** — trace → predict → act → protect, with the six agents and the data sources.
- **Architecture** — the two-seam architecture diagram (schema seam, API seam) — the same SVG the pitch deck uses.
- **Platform** — what runs where (Vercel, Render, Supabase, GitHub Actions) and the ₹0 point.

- **Validation** — the claim / result / how-it-was-checked table: attribution vs published apportionment (cosine 0.88 / 0.90 / 0.93 plus bucket tables), forecast skill vs persistence, the calibrated probabilities, the rejected TFT model, test counts.

VALIDATION		
Every number is checked — including the failures.		
Each claim below is reproducible from the repository — the evaluation notebook and <code>python -m ml.eval.benchmark</code> , whose artifacts the API serves and the console prints. Where a method underperformed, that result ships too.		
CLAIM	RESULT	METHOD
Attribution checked against published apportionment	cosine 0.88 / 0.90 / 0.93	vs SAFAR-Delhi 2018, CSTEP-Bengaluru 2022 (verified from the report), Urban-Emissions Mumbai — plus bucket-by-bucket tables vs TERI-ARAI and Guttikunda et al.
Attribution behaves physically	2.30x traffic signal in rush hours	IST rush vs off-peak SHAP, weather controlled
Forecast beats real baselines	Delhi +9 / +13 / +12% · Mumbai +17 / +19 / +21% at 24/48/72 h	multi-season temporal split (2025-26 winter + summer 2026), monthly refit on the trailing 90 d; served forecast = LightGBM blended with persistence; vs persistence and weekly seasonal-naïve — the raw model's weaker numbers ship too
It warns before the air turns	51-54% of Very-Poor onsets flagged 1-3 days ahead	alarm on the calibrated probability $P \geq 0.3$ (precision 0.64-0.68); persistence catches 0% by construction; the Severe tail stays weak — stated, not hidden
Uncertainty is calibrated	80% band → 78% measured · $P(>120)$ Brier skill +51%	conformal calibration; exceedance probabilities on every cell forecast drive the alarms
Enforcement is equitable	no socio-economic inputs, by construction	fairness audit on every live recommendation (n=390 at the July audit)
Model choice was earned	TFT trained on GPU — and rejected	LightGBM won every launch city on held-out skill
The loop is fast	seconds from signal to cited recommendation	live per-node agent traces; approval, dispatch and closure timestamped per action

4. The console (/console) — the shell

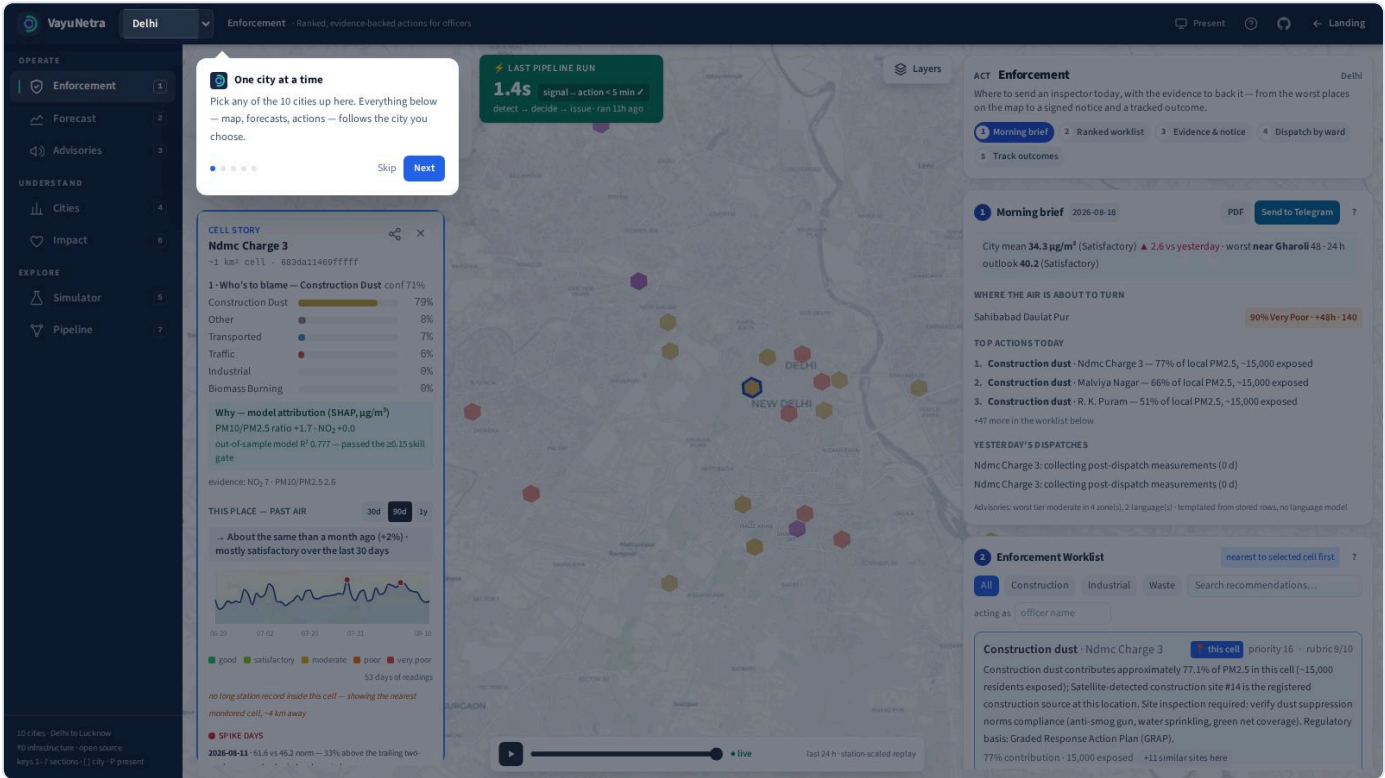


The console is one screen with three columns: the **sidebar** (left, dark), the **map** (centre), and the **rail** (right) that holds the numbered cards of the current section. Every panel follows the same grammar so a new user can predict where things are.

4.1 Top bar

- **VayuNetra** logo → back to the landing page.
- **City selector** — the ten cities. Everything below (map, brief, worklist, forecast, advisories) follows the chosen city; [and] cycle.
- Section title and hint for orientation.
- **Present** — presentation mode: larger type and spacing for a projector (**P**).
- **?** — replays the guided tour.
- **GitHub** — the source.
- **← Landing**.

4.2 Guided tour (first run)



On first visit a four-step tour points at the city selector, the map, the section rail and the spine. It shows once (stored in the browser) and can be replayed from **?**.

4.3 Status strip (top-left of the map)

- **AQI tile** — the city's worst-cell AQI (colour = CPCB band) with the PM2.5 value, the age of the data and a **LIVE** dot.
- **GRAP Stage · from forecast** — the Graded Response Action Plan stage the forecast would trigger (Delhi-NCR only; other cities show their own instrument).
- **Dust × Traffic · N cells** — the dominant-source signature summary.
- **Last pipeline run** — the signal → recommendation compute time of the latest agent run and when it ran.

4.4 Sidebar — seven sections in three groups

Group	Section (key)	One line
Operate	Enforcement (1)	Where to send an inspector today, with the evidence — the officer loop
	Forecast (2)	What the air will do in 72 h, how much to trust it, who it affects, what happened before
	Advisories (3)	What citizens are told, in their language, over which channel

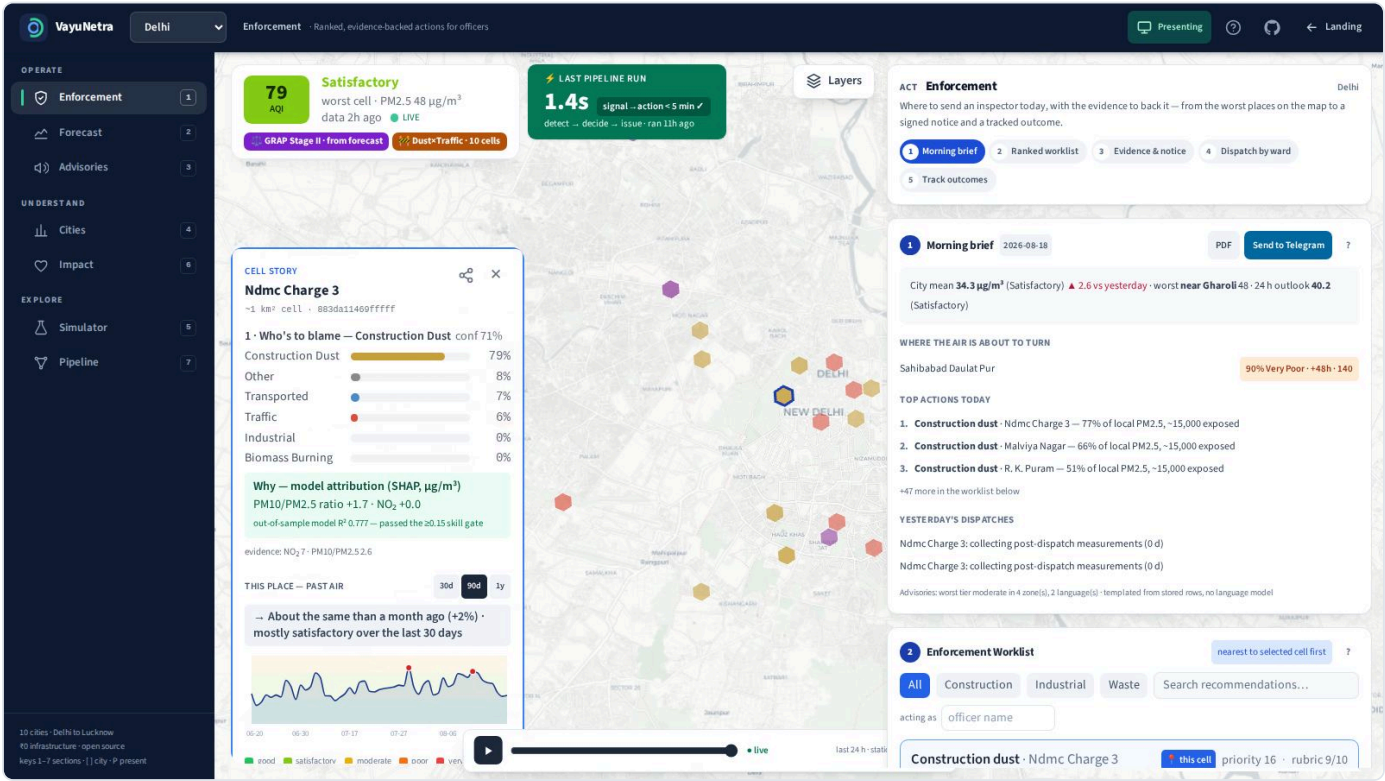
Group	Section (key)	One line
Understand	Cities (4)	Ten cities on one scoreboard
	Impact (6)	The funding case: ₹, lives, where NCAP money should go, fairness
Explore	Simulator (5)	What-if interventions and the inspector-hour optimiser
	Pipeline (7)	The six agents, visibly running

The footer of the sidebar reminds you: ten cities · ₹0 infrastructure · keys 1–7 · [] city · P present.

4.5 The section spine — how anyone knows what to do

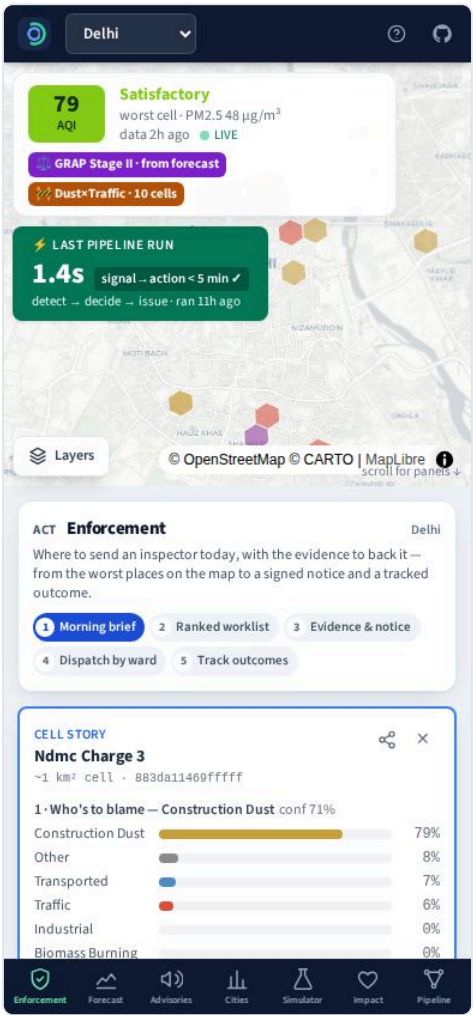
Every section opens with a sticky **spine** at the top of the rail: a verb (*Act, Anticipate, Inform, Compare, Decide, Fund, Trust*), a one-line blurb, and **numbered steps**. Clicking a step scrolls the rail to that card; scrolling highlights the step you are on. Each card carries the same number in its corner and an **?** tip that explains what the card shows, where the data comes from, and what to do next. Steps that live inside another card (e.g. *Evidence & notice* inside the *worklist*) scroll to their parent.

4.6 Presentation mode



Present (or **P**) enlarges type and spacing for a hall projector without changing content.

4.7 Mobile

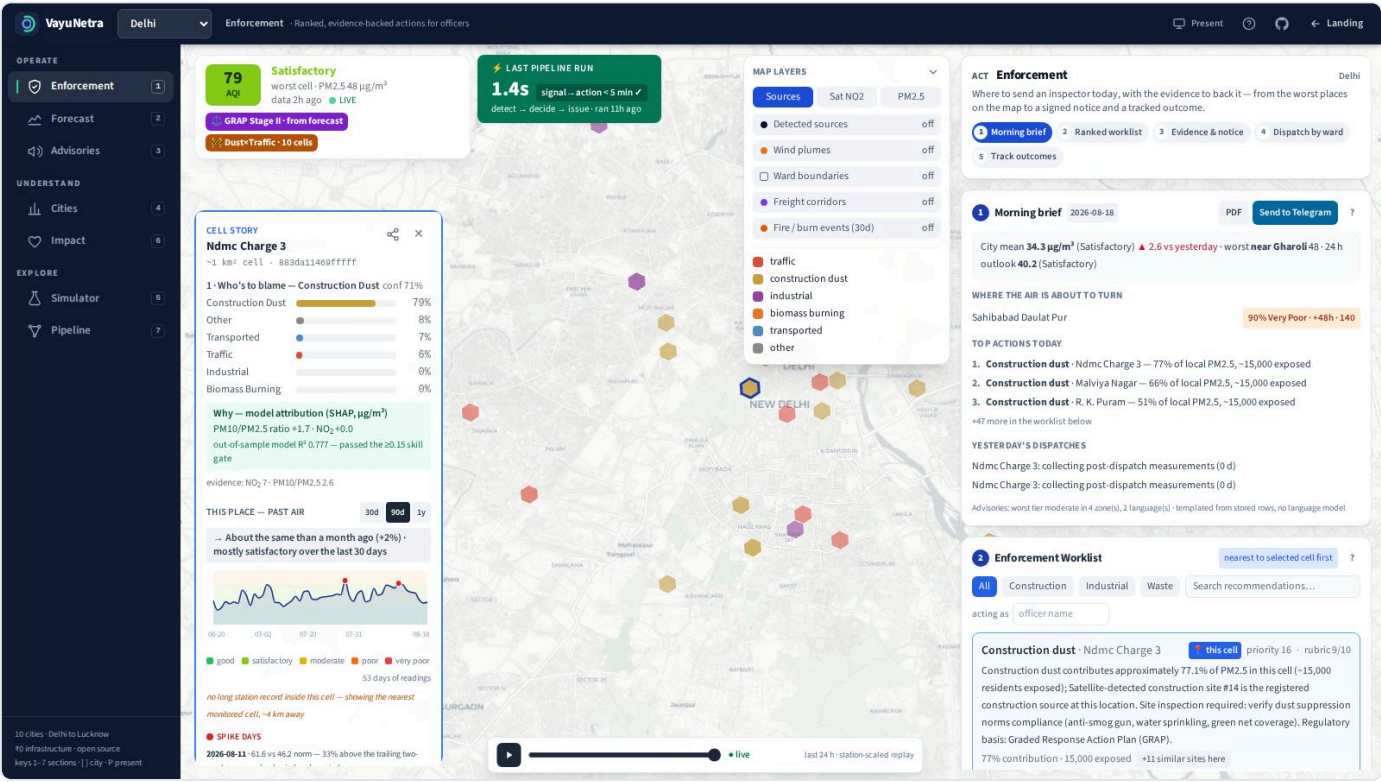


Below 1024 px the rail stacks under the map, the sidebar becomes a bottom navigation, and every tap target is ≥ 24 px. No horizontal scrolling. (Checked by `web/scripts/qa/mobile-check.mjs` ; results in `docs/qa/` .)

5. The map

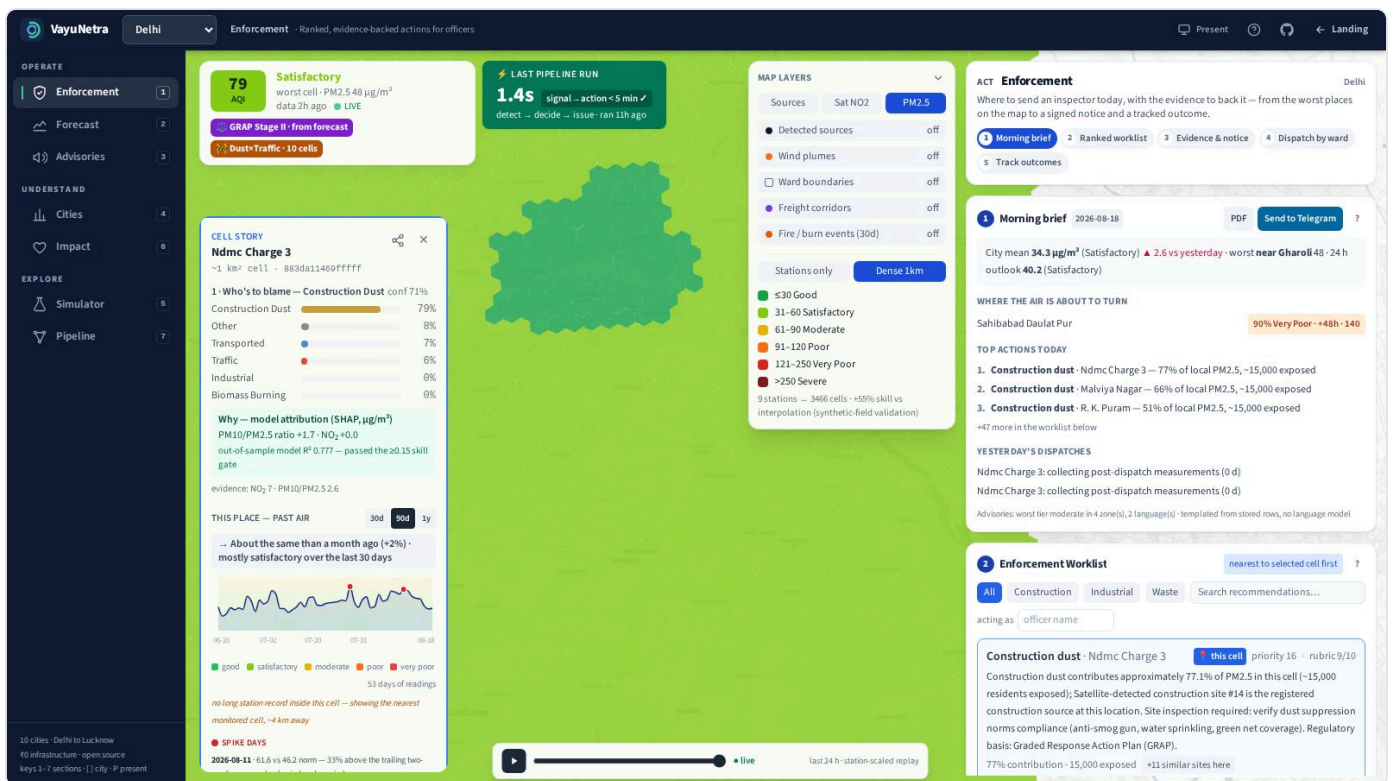
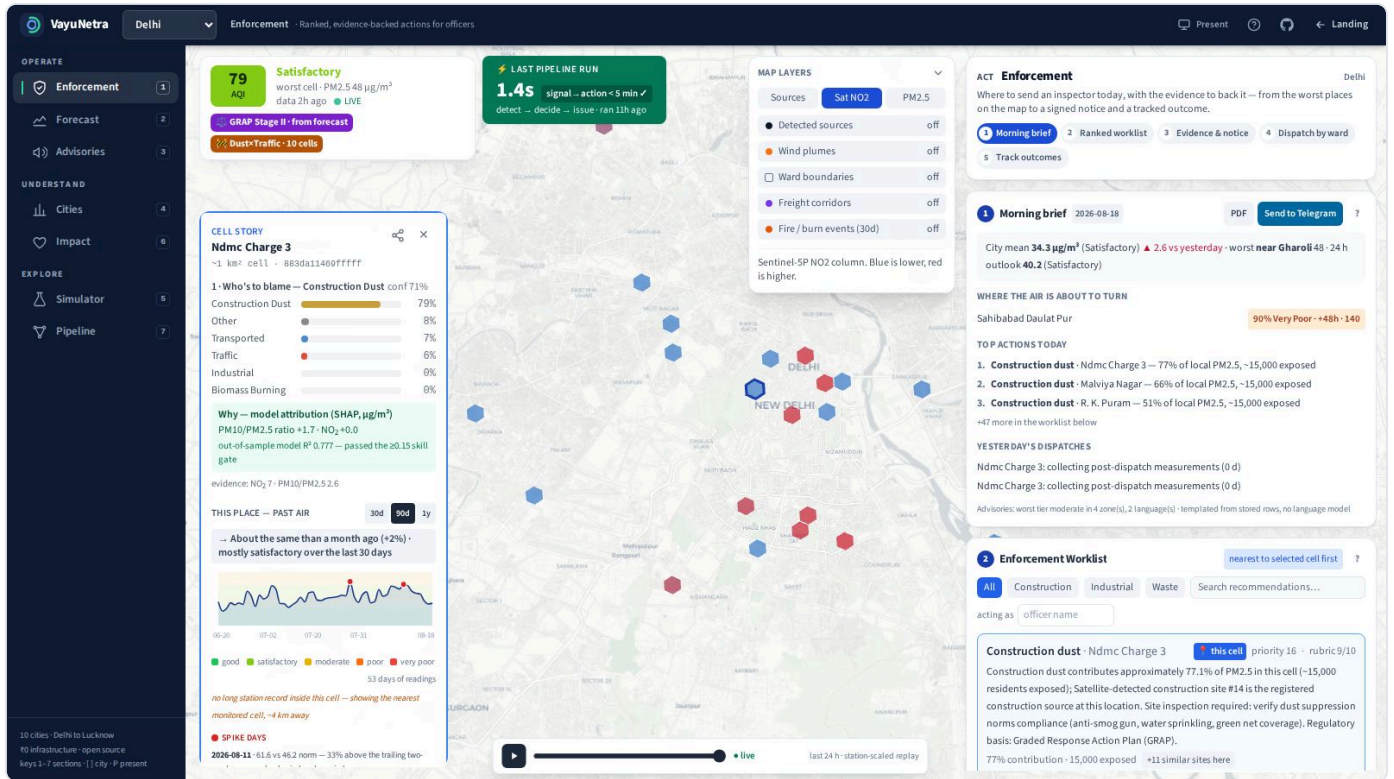
The map is a MapLibre + deck.gl view of the chosen city; every hexagon is one H3 resolution-8 cell (~1 km²).

5.1 The Layers chip — three view modes

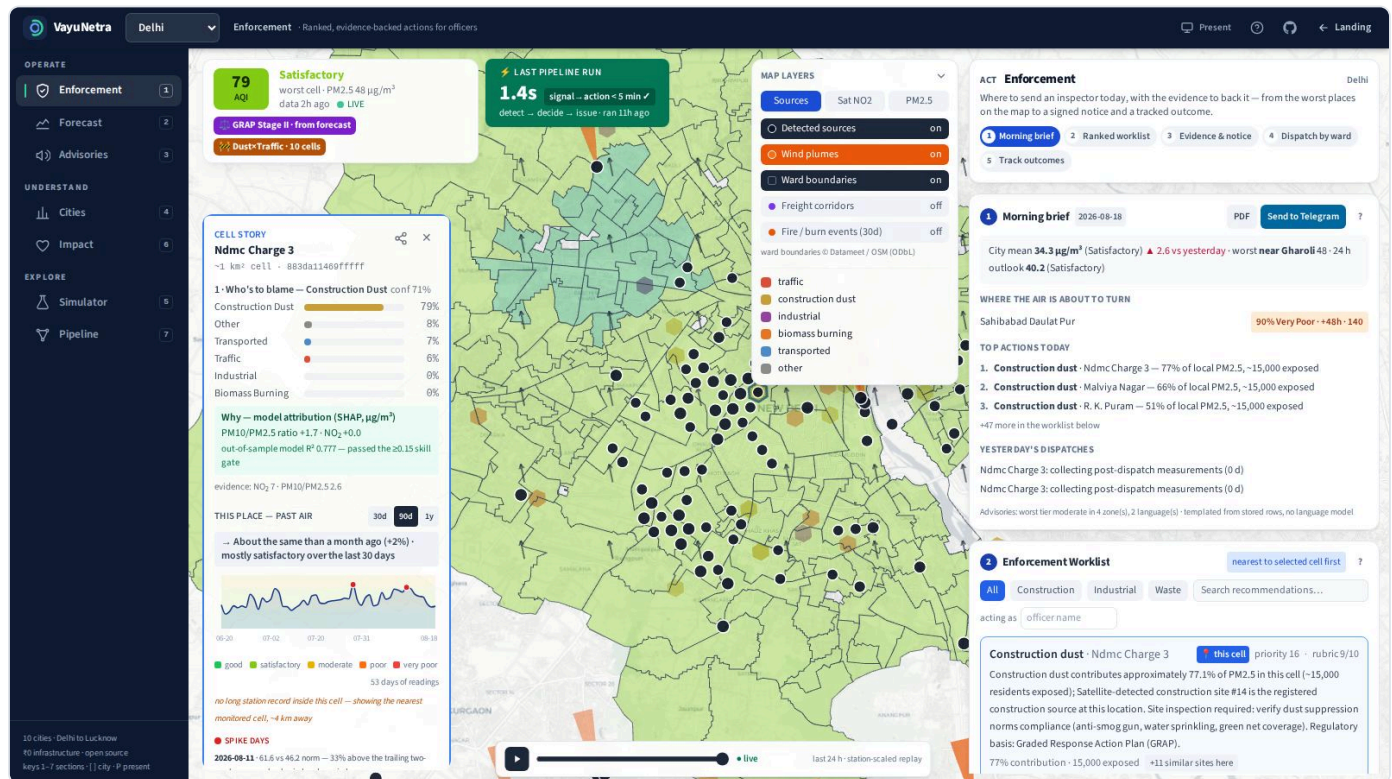


Click **Layers** (top-right of the map) to expand the panel; ▲ collapses it back to a chip.

Mode	What the hexagons show
Sources (default)	Who is to blame — each cell filled by its dominant source colour: traffic red · construction dust yellow · industrial purple · biomass burning orange · transported blue · other grey; opacity = dominance. Hover shows the top-3 shares and SHAP drivers.
Sat NO2	Sentinel-5P tropospheric NO ₂ column (blue low → red high) — independent satellite evidence of combustion.
PM2.5	The dense ~1 km model field (E2 downscaler anchored on the live stations). Sub-toggle Stations only ↔ Dense 1 km shows what the model adds beyond the sparse station grid. Legend = CPCB bands.

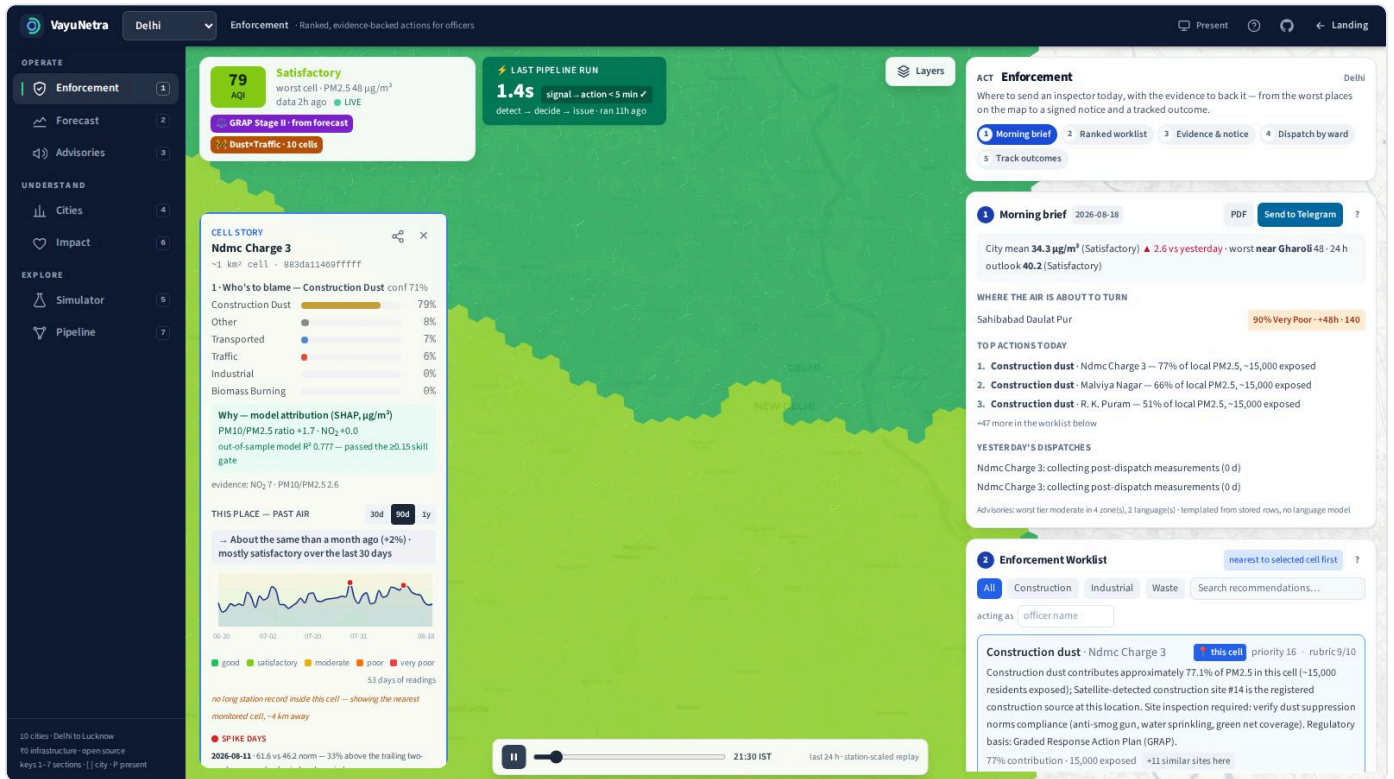


5.2 Five overlays (independent, all off by default)



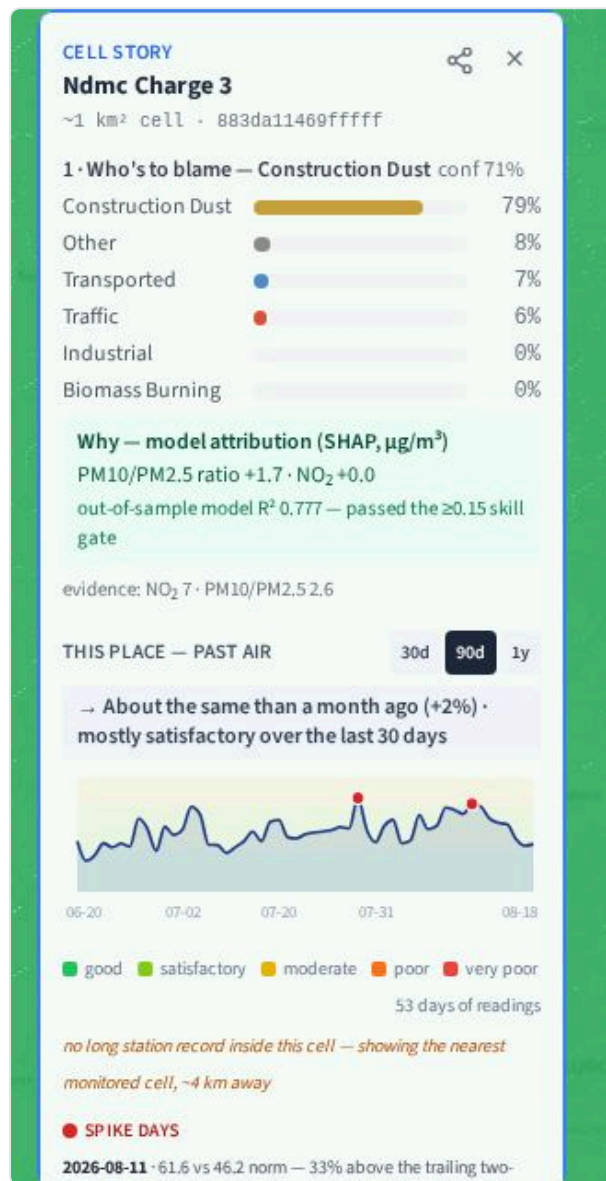
- **Detected sources** — dots for the registered (OSM) and satellite-detected emission sources; **hover a dot to see its real Sentinel-2 image** (sources without one show none — never a stand-in).
- **Wind plumes** — Gaussian-plume footprints from the top sources under the live wind, with a wind-arrow grid (direction; length = speed).
- **Ward boundaries** — administrative wards, each filled by its mean PM2.5 from the dense field (value on hover). The same polygons give every place its name.
- **Freight corridors** — violet truck-route lines with entry-window notes (e.g. Delhi 23:00–07:00).
- **Fire / burn events (30 d)** — NASA FIRMS thermal detections — the stubble / waste-burning layer.

5.3 Play the last 24 hours



The ► control at the bottom of the map replays the trailing day hour by hour so the city visibly breathes through the evening peak and the morning lull; drag the slider to scrub; it returns to **live** at the end. Labelled honestly: *station-scaled replay* (each cell scaled by its nearest monitor's hourly ratio).

5.4 The Cell Story (click any hexagon)



The heart of the console. It opens on load for the worst cell and for any hexagon you click.

1. **Header** — the place name (ward), the raw H3 id, a **share** button (↗) and close (×).
2. **1 · Who's to blame** — the source-share bars with confidence, then one of two *why* boxes: **green — model attribution**: SHAP drivers in µg/m³ and the model's own out-of-sample R² ("passed the ≥ 0.15 skill gate"); **amber — chemical signature**: where the local model lacks skill the system says so and falls back to cited marker-chemistry priors. It never over-claims.
3. **This place — past air** — daily PM2.5 over 30 d / 90 d / 1 y drawn on the CPCB bands, a one-line verdict, and red **spike-day** markers with a data-backed reason. Cells without a long station record say which monitored cell they are borrowing and how far away it is.
4. **2 · Where it's heading** — the cell's +24/48/72 h PM2.5 with 80 % bands, coloured by AQI band, and the **calibrated probability chips** ("63 % Very Poor+", "91 % Severe").
5. **3 · Act — view enforcement actions** → — jumps to the worklist sorted nearest-to-this-cell first.
6. **Share** — renders a WhatsApp-ready PNG card of this place (blame, verdict, 90-day trend, forecast); uses the phone share sheet, else downloads.



6. Enforcement — the officer loop (section 1)

Where to send an inspector today, with the evidence to back it — from the worst places on the map to a signed notice and a tracked outcome. Steps: **1 Morning brief** · **2 Ranked worklist** · **3 Evidence & notice** · **4 Dispatch by ward** · **5 Track outcomes**.

6.1 Morning brief (step 1)

1 Morning brief2026-08-18PDFSend to Telegram?

City mean 34.3 $\mu\text{g}/\text{m}^3$ (Satisfactory) ▲ 2.6 vs yesterday · worst near Gharoli 48 · 24 h outlook 40.2 (Satisfactory)

WHERE THE AIR IS ABOUT TO TURN

Sahibabad Daulat Pur90% Very Poor · +48h · 140

TOP ACTIONS TODAY

1. Construction dust · Ndmc Charge 3 — 77% of local PM2.5, ~15,000 exposed

2. Construction dust · Malviya Nagar — 66% of local PM2.5, ~15,000 exposed

3. Construction dust · R. K. Puram — 51% of local PM2.5, ~15,000 exposed

+47 more in the worklist below

YESTERDAY'S DISPATCHES

Ndmc Charge 3: collecting post-dispatch measurements (0 d)

Ndmc Charge 3: collecting post-dispatch measurements (0 d)

Advisories: worst tier moderate in 4 zone(s), 2 language(s) · templated from stored rows, no language model

2 Enforcement Worklist?

AllConstructionIndustrialWasteSearch recommendations...

acting asofficer name

Construction dust · Ndmc Charge 3priority 16 · rubric 9/10

Construction dust contributes approximately 77.1% of PM2.5 in this cell (~15,000 residents exposed); Satellite-detected construction site #14 is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).

77% contribution · 15,000 exposed+11 similar sites here

The one page a commissioner reads. Every line is a template over stored rows — no language model: - **Air right now** — city mean PM2.5 (band) vs the same hours yesterday (▲/▼), the worst place, the 24 h outlook; if the last reading is old, its age is printed. - **Where the air is about to turn** — cells whose calibrated P(Very Poor) $\geq 30\%$ at any horizon and that are not already Very Poor, with the horizon and forecast value. - **Top actions today** — the three highest-priority open recommendations (source · place — contribution, residents exposed) with a "+N more" line. - **Yesterday's dispatches** — each dispatched action's measured effect vs the city's drift, or the honest empty state. - **Advisories** — worst tier issued, in how many zones and languages. - **PDF** downloads the brief; **Send to Telegram** pushes it to the city's subscribers (real send; the API reports "skipped" honestly if no bot token is configured). The nightly pipeline sends the brief automatically.

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6.2 Enforcement workflow (step 2)

1

Morning brief

2026-08-18

PDF

Send to Telegram

?

City mean **34.3 $\mu\text{g}/\text{m}^3$** (Satisfactory) ▲ 2.6 vs yesterday · worst **near Gharoli** 48 · 24 h outlook **40.2** (Satisfactory)

WHERE THE AIR IS ABOUT TO TURN

Sahibabad Daulat Pur 90% Very Poor · +48h · 140

TOP ACTIONS TODAY

1. **Construction dust** · Ndmc Charge 3 — 77% of local PM2.5, ~15,000 exposed

2. **Construction dust** · Malviya Nagar — 66% of local PM2.5, ~15,000 exposed

3. **Construction dust** · R. K. Puram — 51% of local PM2.5, ~15,000 exposed

+47 more in the worklist below

YESTERDAY'S DISPATCHES

Ndmc Charge 3: collecting post-dispatch measurements (0 d)

Ndmc Charge 3: collecting post-dispatch measurements (0 d)

Advisories: worst tier moderate in 4 zone(s), 2 language(s) · templated from stored rows, no language model

2

Enforcement Worklist

?

All

Construction

Industrial

Waste

Search recommendations...

acting as

Construction dust · Ndmc Charge 3

priority 16 · rubric 9/10

Construction dust contributes approximately 77.1% of PM2.5 in this cell (~15,000 residents exposed); Satellite-detected construction site #14 is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).

77% contribution · 15,000 exposed +11 similar sites here

Ranked recommendations, each titled **source type · place** with its priority and rubric score. Filters **All / Construction / Industrial / Waste**, free-text **search**, and the **acting as** field (the officer name stamped on every action; the demo has no sign-in — a deployment binds this to the authenticated user). If a cell is focused, sorting switches to *nearest first* (badge + "~N km" / "📍 this cell" chips). Identical detections in one cell collapse into one card (" +N similar sites here").

Each card: the rationale (what the source is, its % contribution, residents exposed, what to inspect, the regulatory basis), then the buttons **Evidence dossier · Notice PDF · Approve · Dismiss**.

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6.3 Evidence dossier and notice (step 3)

Where to send an inspector today, with the evidence to back it — from the worst places on the map to a signed notice and a tracked outcome.

1 Morning brief 2 **Ranked workload** 3 Evidence & notice 4 Dispatch by ward 5 Track outcomes

acting as officer name

Construction dust · Ndmc Charge 3 priority 16 · rubric 9/10

Construction dust contributes approximately 77.1% of PM2.5 in this cell (~15,000 residents exposed); Satellite-detected construction site #14 is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).

77% contribution · 15,000 exposed +11 similar sites here

Hide dossier Notice PDF Approve Dismiss

SATELLITE EVIDENCE



Sentinel-2 patch - Satellite-detected construction site #14
construction · 90% detection confidence

REGULATORY CITATIONS (RAG)

Graded Response Action Plan (GRAP) match 24%

hospitals: additional respiratory care wards activated. - Helpline numbers published; mobile health units deployed.

- **Evidence dossier** expands in place: the **Sentinel-2 satellite patch** of the site (with detection confidence) and up to three **regulatory citations** retrieved by RAG from the NCAP / GRAP / CPCB / Air-Act corpus (GRAP/CAQM only where they legally bind — Delhi-NCR; CPCB dust norms and NCAP elsewhere), each with a match score.
- **Notice PDF** downloads a draft enforcement notice: addressee, cited provisions, 24-hour IST deadline, the satellite image, a **projected impact of compliance** chart (forecast with vs without this source's share at +24/48/72 h — a screening estimate), the issuing authority, and a **PROVENANCE** block. Stamped *DRAFT — pending officer authorisation*; the system never auto-sends. Live assembly takes 5–12 s.

6.4 Approve → Dispatch → Close → History

acting as Guide waikarouhi

Construction dust · Ndmc Charge 3 priority 16 · rubric 9/10

Construction dust contributes approximately 77.1% of PM2.5 in this cell (~15,000 residents exposed); Satellite-detected construction site #14 is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).

77% contribution · 15,000 exposed +11 similar sites here

Evidence dossier Notice PDF Dispatch team History

Construction dust · Malviya Nagar priority 12 · rubric 8/10

Construction dust contributes approximately 66.1% of PM2.5 in this cell (~15,000 residents exposed); Satellite-detected construction site #39 is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).

66% contribution · 15,000 exposed +3 similar sites here

Evidence dossier Notice PDF Approve Dismiss

Construction dust · R. K. Puram priority 9 · rubric 8/10

Construction dust contributes approximately 51.1% of PM2.5 in this cell (~15,000 residents exposed); Satellite-detected construction site #28 is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).

51% contribution · 15,000 exposed +7 similar sites here

Evidence dossier Notice PDF Approve Dismiss

Construction dust · near Madanpur Khadar priority 6 · rubric 8/10

- **Approve** moves the item to its ward's field queue (step 4). **Dismiss** removes it.
- **Dispatch team** freezes the cell's 7-day PM2.5 baseline and **arms before/after tracking** (step 5); the card shows *Dispatched · tracking armed*.

acting as [Guide walkthrough](#)

Construction dust · Ndmc Charge 3
priority 16 · rubric 9/10

Construction dust contributes approximately 77.1% of PM2.5 in this cell (~15,000 residents exposed); Satellite-detected construction site #14 is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).

77% contribution · 15,000 exposed [+11 similar sites here](#)

[Evidence dossier](#)
[Notice PDF](#)
[✓ Dispatched · tracking armed](#)

[Close case](#)
[History](#)

Field finding
violation found ▼
note (optional): what
[Record & close](#)

Construction dust · Malviya Nagar
priority 12 · rubric 8/10

Construction dust contributes approximately 66.1% of PM2.5 in this cell (~15,000 residents exposed); Satellite-detected construction site #39 is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).

66% contribution · 15,000 exposed [+3 similar sites here](#)

[Evidence dossier](#)
[Notice PDF](#)
[Approve](#)
[Dismiss](#)

Construction dust · R. K. Puram
priority 9 · rubric 8/10

Construction dust contributes approximately 51.1% of PM2.5 in this cell (~15,000 residents exposed); Satellite-detected construction site #28 is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).

51% contribution · 15,000 exposed [+7 similar sites here](#)

- **Close case** records the field finding — *violation found · compliant · site inaccessible · not applicable* — and an optional note; **Record & close** stores it and the card shows *Closed · finding · date*.

acting as Guide walkthrough

Construction dust · Ndmc Charge 3 priority 16 · rubric 9/10

Construction dust contributes approximately 77.1% of PM2.5 in this cell (~15,000 residents exposed); Satellite-detected construction site #14 is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).

77% contribution · 15,000 exposed +11 similar sites here

Evidence dossier Notice PDF ✓ Closed · not applicable · 2026-08-18

Hide history

18 Aug, 11:53 am proposed → approved
 18 Aug, 11:53 am approved → dispatched
 18 Aug, 11:53 am dispatched → closed by Journey test not applicable
 "e2e journey — no field visit"
 18 Aug, 11:56 am closed → proposed by QA reset "reset after e2e run"
 18 Aug, 07:28 pm proposed → approved by Guide walkthrough
 18 Aug, 07:28 pm approved → dispatched by Guide walkthrough
 18 Aug, 07:28 pm dispatched → closed by Guide walkthrough not applicable
 "user-guide walkthrough — no field visit"
 18 Aug, 07:33 pm closed → proposed by Guide walkthrough
 "reset after walkthrough"
 18 Aug, 07:52 pm proposed → approved by Guide walkthrough
 18 Aug, 07:52 pm approved → dispatched by Guide walkthrough
 18 Aug, 07:52 pm dispatched → closed by Guide walkthrough not applicable
 "user-guide walkthrough — no field visit"

Construction dust · Malviya Nagar priority 12 · rubric 8/10

Construction dust contributes approximately 66.1% of PM2.5 in this cell (~15,000 residents exposed); Satellite-detected construction site #20 is the registered

- **History** shows the audit trail: from → to, who (acting as), when, finding, note. It is append-only (enforcement_status_log), survives the nightly run — which refreshes evidence and priority *in place* for acted-upon items and replaces only still-proposed ones — and survives even deletion of the record.

6.5 Ward dispatch queues (step 4)

Where to send an inspector today, with the evidence to back it — from the worst places on the map to a signed notice and a tracked outcome.

1 Morning brief 2 Ranked worklist 3 Evidence & notice 4 Dispatch by ward 5 Track outcomes

Construction dust · Ward PUSA priority 3 · rubric 8/10

Construction dust contributes approximately 34.3% of PM2.5 in this cell (~15,000 residents exposed); Satellite-detected construction site #20 is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).

34% contribution · 15,000 exposed [+6 similar sites here](#)

[Evidence dossier](#) [Notice PDF](#) [Approve](#) [Dismiss](#)

4 Ward dispatch queues EMPTY ?

Nothing dispatched yet. Approve or dispatch a worklist item above and it lands in its ward's field queue here.

5 Intervention tracking ?

rec #12902 · cell 9ffff measuring

measuring — 0 days since dispatch · collecting post-dispatch measurements

rec #12901 · cell 9ffff measuring

measuring — 0 days since dispatch · collecting post-dispatch measurements

Export as NCAP action-plan evidence (PRANA-ready CSV) ↓ — each dispatched intervention with its measured effect, mapped to the NCAP spending head the city reports against. We feed the official portal, not compete with it.

Approved and dispatched recommendations group into per-ward field queues (the grid-supervision pattern); wards are resolved offline from the shipped boundary polygons. Before anything is dispatched the card says so plainly.

6.6 Intervention tracking and PRANA export (step 5)

Where to send an inspector today, with the evidence to back it — from the worst places on the map to a signed notice and a tracked outcome.

1 Morning brief

2 Ranked worklist

3 Evidence & notice

4 Dispatch by ward

5 Track outcomes

Construction dust · Ward PUSA

priority 3 · rubric 8/10

Construction dust contributes approximately 34.3% of PM2.5 in this cell (~15,000 residents exposed); Satellite-detected construction site #20 is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).

34% contribution · 15,000 exposed

+6 similar sites here

Evidence dossier

Notice PDF

Approve

Dismiss

4 Ward dispatch queues

EMPTY

?

Nothing dispatched yet. Approve or dispatch a worklist item above and it lands in its ward's field queue here.

5 Intervention tracking

?

rec #12902 · cell 9ffff

measuring

measuring — 0 days since dispatch · collecting post-dispatch measurements

rec #12901 · cell 9ffff

measuring

measuring — 0 days since dispatch · collecting post-dispatch measurements

Export as NCAP action-plan evidence (PRANA-ready CSV) ↓ — each dispatched intervention with its measured effect, mapped to the NCAP spending head the city reports against. We feed the official portal, not compete with it.

Each dispatched action's measured effect: the cell's PM2.5 after dispatch minus its frozen baseline, corrected for the city-wide drift over the same window (marked *provisional* for the first 7 days). **Export as NCAP action-plan evidence** downloads the PRANA-ready CSV — every tracked action with baseline / after / effect / status / closure finding, mapped to the NCAP spending head the city reports against.

6.7 City Intel

Where to send an inspector today, with the evidence to back it — from the worst places on the map to a signed notice and a tracked outcome.

1 Morning brief

2 Ranked worklist

3 Evidence & notice

4 Dispatch by ward

5 Track outcomes

Construction dust · Ward PUSA

priority 3 · rubric 8/10

Construction dust contributes approximately 34.3% of PM2.5 in this cell (~15,000 residents exposed); Satellite-detected construction site #20 is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).

34% contribution · 15,000 exposed

+6 similar sites here

Evidence dossier

Notice PDF

Approve

Dismiss

4 Ward dispatch queues

EMPTY

?

Nothing dispatched yet. Approve or dispatch a worklist item above and it lands in its ward's field queue here.

5 Intervention tracking

?

rec #12902 · cell 9ffff

measuring

measuring — 0 days since dispatch · collecting post-dispatch measurements

rec #12901 · cell 9ffff

measuring

measuring — 0 days since dispatch · collecting post-dispatch measurements

Export as NCAP action-plan evidence (PRANA-ready CSV) ↕ — each dispatched intervention with its measured effect, mapped to the NCAP spending head the city reports against. We feed the official portal, not compete with it.

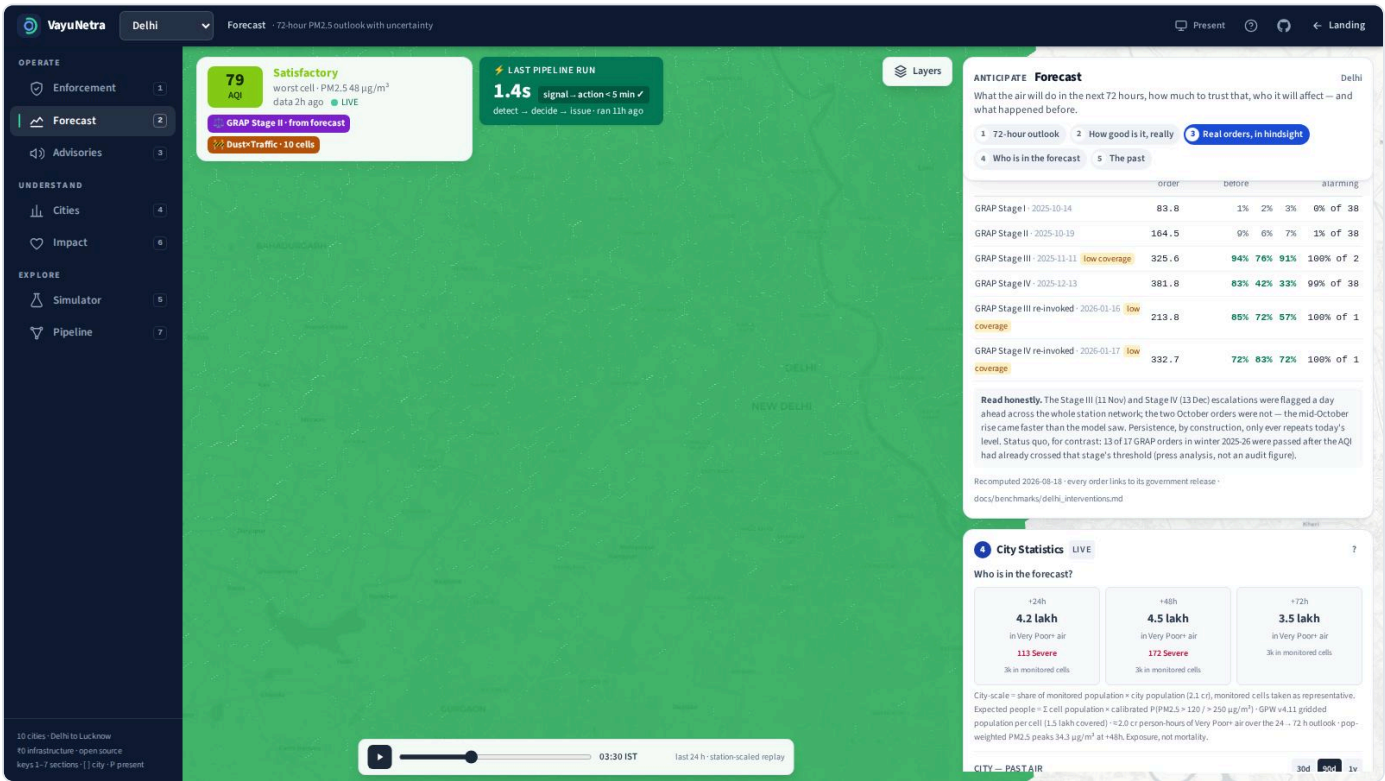
City Intel

Registry-source count, traffic index, sensitive-zone count, hospitals / schools, top registered sources — the static picture behind the live one.

7. Forecast (section 2)

What the air will do in the next 72 hours, how much to trust that, who it will affect — and what happened before. Steps: 1 72-hour outlook · 2 How good is it, really · 3 Real orders, in hindsight · 4 Who is in the forecast · 5 The past.

7.1 72-hour outlook



Horizon tabs **+24h / +48h / +72h**; the city-average forecast line with the shaded **80 % band** and the grey dashed **persistence** baseline ("tomorrow = today" — the bar to beat). The **measured skill note** under the chart is read from the benchmark artifact, never typed in. **Spike alerts** list cells forecast $\geq 90 \mu\text{g}/\text{m}^3$ (worst first). Model: LightGBM quantile regression, retrained daily per city on the trailing 90 days, blended with persistence, conformally calibrated; every cell carries $P(>120)$ and $P(>250)$.

7.2 Forecast validation (step 2)



The strict temporal-split benchmark: skill vs persistence and weekly seasonal-naive per horizon, winter-only and Very-Poor-hours slices, 80 % interval coverage, and the **early-warning line** — how many clean→Very-Poor onsets the alarm on $P \geq 0.3$ flags 1–3 days ahead (persistence = 0 by construction). Toggle **multi-season / live 90d**. Negative numbers stay visible. Below it, **How today's attribution was made** — cells per method (per-cell model · shrunk to the city model mean · signature priors), median R^2 , mean confidence, cells with a gas marker in the last 24 h. Sources: docs/BENCHMARKS.md , docs/EARLY_WARNING.md , GET /metrics/benchmark , GET /metrics/attribution .

7.3 Real interventions, in hindsight (step 3)

What the air will do in the next 72 hours, how much to trust that, who it will affect — and what happened before.

1 72-hour outlook

2 How good is it, really

3 Real orders, in hindsight

4 Who is in the forecast

5 The past

calibrated probability ($P \geq 30\%$) flags 54% @24h · 54% @48h · 51% @72h of onsets (precision 68% · 66% · 64%); the plain median alarm 24% · 31% · 27%. Persistence catches 0% by construction — a "tomorrow = today" system can never warn of a spike before it starts.

Recomputed 2026-08-17 · negative numbers are kept, not hidden · full tables in docs/benchmarks/delhi.md

How today's attribution was made

24 cells

shrunk to city model mean 21

per-cell model 3

median out-of-sample R^2 0.78 · mean confidence 0.39 · cells reporting $\text{NO}_2/\text{CO}/\text{SO}_2$ in the last 24 h 2

recomputed daily from the latest hour; the split moves with station marker coverage — stated, not smoothed

3 Real int...

WINTER 2025-26

would we have warned?

did the air change?

?

CAQM's GRAP orders for Delhi-NCR, winter 2025-26 (dates from the government releases), against the served forecast replayed on 39 station cells; the calibrated $P(>120)$ the system carried 24 / 48 / 72 h before each order, and the share of cells past the $P \geq 30\%$ alarm.

order	at order	$P(>120)$ 24 h before	48 h	72 h	cells alarming
GRAP Stage I · 2025-10-14	83.8	1%	2%	3%	0% of 38
GRAP Stage II · 2025-10-19	164.5	9%	6%	7%	1% of 38
GRAP Stage III · 2025-11-11 low coverage	325.6	94%	76%	91%	100% of 2
GRAP Stage IV · 2025-12-13	381.8	83%	42%	33%	99% of 38
GRAP Stage III re-invoked · 2026-01-16 low coverage	213.8	85%	72%	57%	100% of 1
GRAP Stage IV re-invoked · 2026-01-17 low coverage	332.7	72%	83%	72%	100% of 1

Read honestly. The Stage III (11 Nov) and Stage IV (13 Dec) escalations were flagged a day ahead across the whole station network; the two October orders were not — the mid-October rise came faster than the model saw. Persistence, by construction, only ever repeats today's level, Status quo, for contrast: 13 of 17 GRAP orders in winter 2025-26 were passed after the AQI

What the air will do in the next 72 hours, how much to trust that, who it will affect — and what happened before.

1 72-hour outlook 2 How good is it, really 3 **Real orders, in hindsight** 4 Who is in the forecast 5 The past

calibrated probability ($P \geq 30\%$) flags 54% @24h · 54% @48h · 51% @72h of onsets (precision 68% · 66% · 64%); the plain median alarm 24% · 31% · 27%. Persistence catches 0% by construction — a "tomorrow = today" system can never warn of a spike before it starts.

Recomputed 2026-08-17 · negative numbers are kept, not hidden · full tables in docs/benchmarks/delhi.md

How today's attribution was made 24 cells

shrunk to city model mean 21 per-cell model 3

median out-of-sample R^2 0.78 · mean confidence 0.39 · cells reporting $\text{NO}_2/\text{CO}/\text{SO}_2$ in the last 24 h 2

recomputed daily from the latest hour; the split moves with station marker coverage — stated, not smoothed

3 **Real int...** WINTER 2025-26 would we have warned? **did the air change?** ?

A city-wide order has no untreated control and is triggered by dirty air, so a plain before/after measures weather and regression to the mean. Instead: what $\text{PM}_{2.5}$ *that weather* normally brings (LightGBM on ERA5 meteorology + calendar, fitted on the season's hours outside the windows; held-out R^2 0.61, RMSE 65 $\mu\text{g}/\text{m}^3$) vs what was observed.

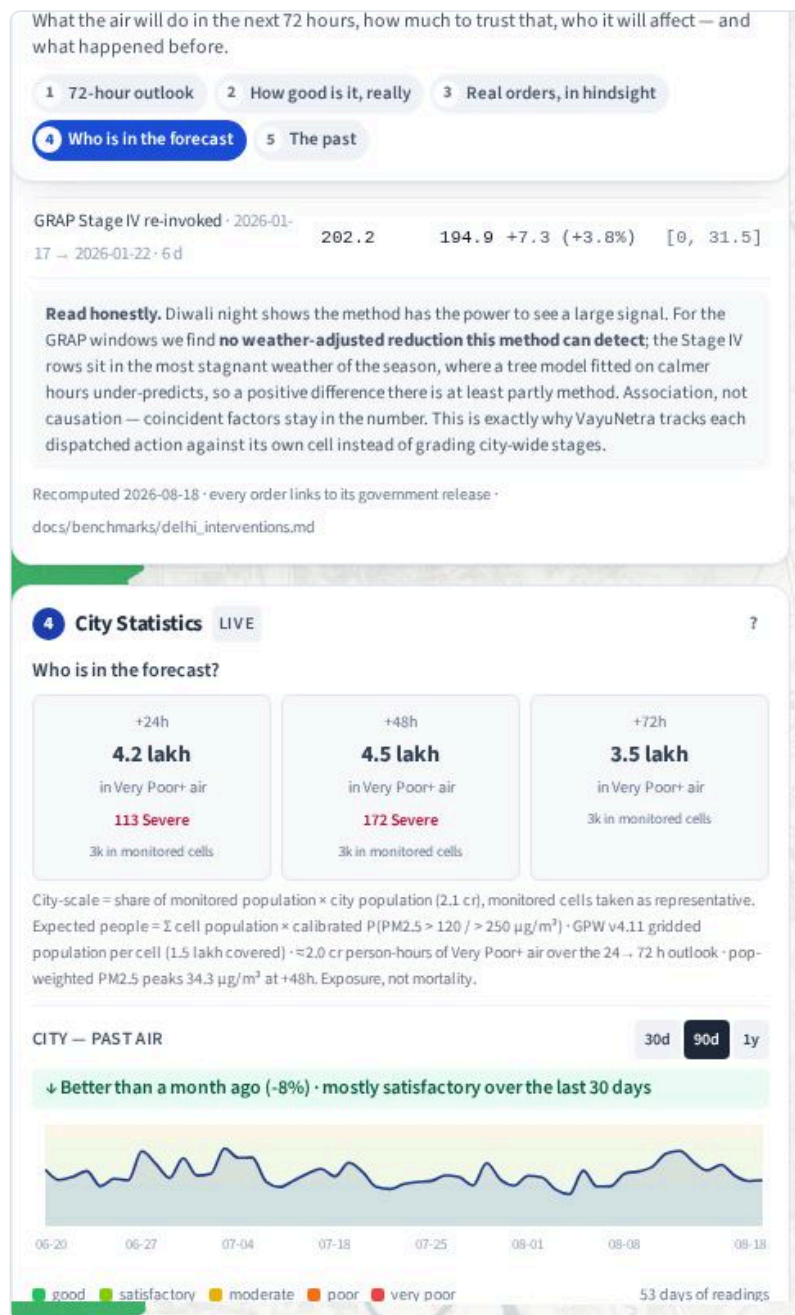
window	observed	weather-expected	difference	90% interval
Diwali night · 2025-10-20 → 2025-10-21 · 2 d	373.4	191.6	+181.8 (+94.9%)	[92.1, 314.4]
GRAP Stage III · 2025-11-11 → 2025-11-26 · 16 d	239.3	239.2	+0.2 (+0.1%)	[-14.9, 14]
GRAP Stage IV · 2025-12-13 → 2025-12-24 · 12 d	241.6	200.5	+41.1 (+20.5%)	[22.6, 58.7]
GRAP Stage III re-invoked · 2026-01-16 → 2026-01-22 · 7 d	203.3	195.8	+7.5 (+3.8%)	[0.1, 32.8]
GRAP Stage IV re-invoked · 2026-01-17 → 2026-01-22 · 6 d	202.2	194.9	+7.3 (+3.8%)	[0, 31.5]

Read honestly. Diwali night shows the method has the power to see a large signal. For the

The CAQM GRAP escalations of winter 2025-26 (and Diwali night), dated from government releases (each order links to its source), replayed against the served forecast: **would we have warned?** shows $P(>120)$ 24/48/72 h before each order and the share of station cells past the alarm (low-coverage weeks flagged); **did the air change?** shows observed vs weather-expected $\text{PM}_{2.5}$ inside each window with a bootstrap interval — where the honest answer is "no detectable change", it says no.

docs/OUTCOMES.md , GET /metrics/interventions .

7.4 City Statistics (steps 4–5)

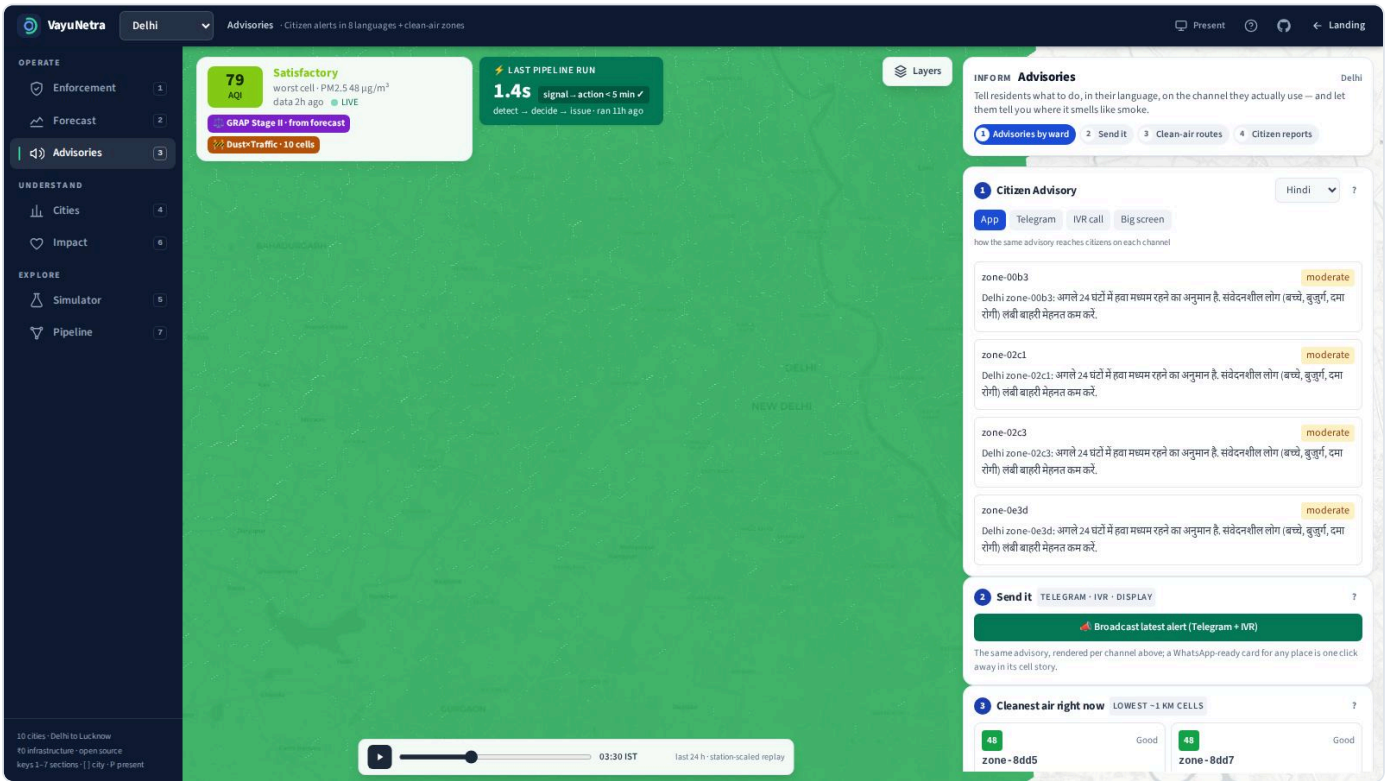


- **Who is in the forecast?** — expected people in Very Poor / Severe air at +24/48/72 h = Σ cell population × calibrated $P(> \text{band})$; GPW v4.11 population where sampled, cited city population otherwise; exposure, not mortality ([docs/HEALTH_IMPACT.md](#)).
- **City — past air** — daily $\text{PM}_{2.5}$ for the whole city over 30 d / 90 d / 1 y with verdict and spike days (raw readings u the archived daily rollup — Delhi's 1-year view shows the real winter).
- **Last 48 hours** — hourly station means; the live **source-mix donut**; **Who breathes what** — share of cells in each CPCB band.

8. Advisories — the citizen loop (section 3)

Inform: what citizens are told, in their language, over which channel. Steps: **1 Advisories by ward** · **2 Send it** · **3 Clean-air routes** · **4 Citizen reports**.

8.1 Citizen Advisory



Tell residents what to do, in their language, on the channel they actually use — and let them tell you where it smells like smoke.

1 **Advisories by ward** 2 Send it 3 Clean-air routes 4 Citizen reports

1 **Citizen Advisory** Hindi ?

App Telegram IVR call Big screen

how the same advisory reaches citizens on each channel

zone-00b3 moderate

Delhi zone-00b3: अगले 24 घंटों में हवा मध्यम रहने का अनुमान है. संवेदनशील लोग (बच्चे, बुजुर्ग, दमा रोगी) लंबी बाहरी मेहनत कम करें.

zone-02c1 moderate

Delhi zone-02c1: अगले 24 घंटों में हवा मध्यम रहने का अनुमान है. संवेदनशील लोग (बच्चे, बुजुर्ग, दमा रोगी) लंबी बाहरी मेहनत कम करें.

zone-02c3 moderate

Delhi zone-02c3: अगले 24 घंटों में हवा मध्यम रहने का अनुमान है. संवेदनशील लोग (बच्चे, बुजुर्ग, दमा रोगी) लंबी बाहरी मेहनत कम करें.

zone-0e3d moderate

Delhi zone-0e3d: अगले 24 घंटों में हवा मध्यम रहने का अनुमान है. संवेदनशील लोग (बच्चे, बुजुर्ग, दमा रोगी) लंबी बाहरी मेहनत कम करें.

2 **Send it** TELEGRAM · IVR · DISPLAY ?

 Broadcast latest alert (Telegram + IVR)

The same advisory, rendered per channel above; a WhatsApp-ready card for any place is one click away in its cell story.

3 **Cleanest air right now** LOWEST ~1 KM CELLS ?

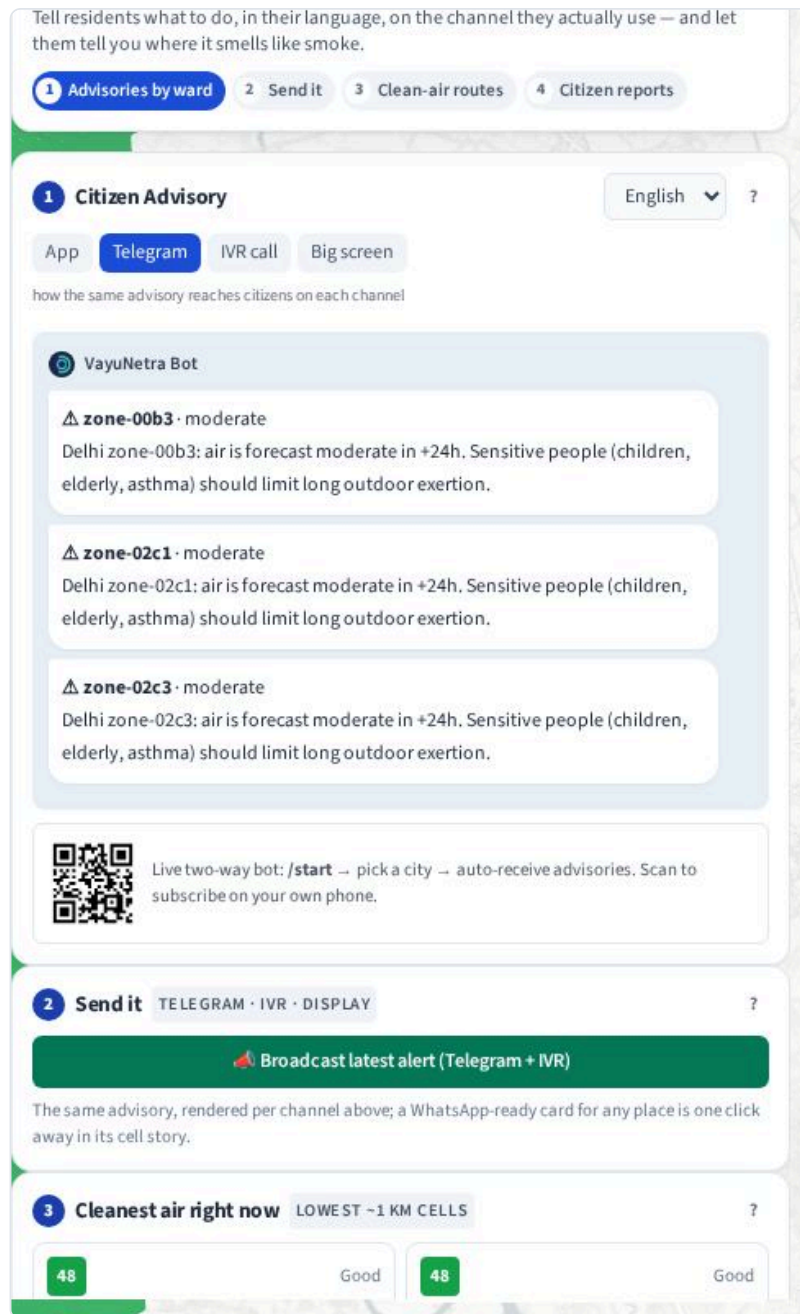
48 Good

zone-8dd5

48 Good

zone-8dd7

Ward-level advisories tiered by forecast risk and vulnerability. **Language dropdown:** the city's showcase language first (Hindi in Delhi, Marathi in Mumbai/Pune, Kannada in Bengaluru...), then the others — full native-script text from deterministic templates (LLM-free by design, so medical advice cannot be hallucinated; script-validated in code; native-speaker review status per language in `docs/ADVISORY_REVIEW.md`). **Channel tabs** show how the same advisory renders on each channel:



- **App** — advisory cards per ward with risk-tier badges.
- **Telegram** — the chat rendering + QR; the real bot **@aqivayu_bot** is live: `/start` → pick a city → receive advisories.
- **IVR call** — the spoken script, plus the call-in line: a keypad menu of the ten cities reads that city's latest advisory — Hindi voice for Hindi-first cities, English elsewhere (no Polly voice exists yet for the other six scripts; those calls read English — stated on screen).
- **Big screen** — high-contrast public-display rendering.

8.2 Send it

Tell residents what to do, in their language, on the channel they actually use — and let them tell you where it smells like smoke.

1 **Advisories by ward** 2 **Send it** 3 **Clean-air routes** 4 **Citizen reports**

1 Citizen Advisory English ?

App Telegram IVR call Big screen

how the same advisory reaches citizens on each channel

zone-00b3 moderate

Delhi zone-00b3: air is forecast moderate in +24h. Sensitive people (children, elderly, asthma) should limit long outdoor exertion.

zone-02c1 moderate

Delhi zone-02c1: air is forecast moderate in +24h. Sensitive people (children, elderly, asthma) should limit long outdoor exertion.

zone-02c3 moderate

Delhi zone-02c3: air is forecast moderate in +24h. Sensitive people (children, elderly, asthma) should limit long outdoor exertion.

zone-0e3d moderate

Delhi zone-0e3d: air is forecast moderate in +24h. Sensitive people (children, elderly, asthma) should limit long outdoor exertion.

2 Send it TELEGRAM · IVR · DISPLAY ?

 **Broadcast latest alert (Telegram + IVR)**

The same advisory, rendered per channel above; a WhatsApp-ready card for any place is one click away in its cell story.

3 Cleanest air right now LOWEST ~1 KM CELLS ?

48 Good

zone-8dd5

48 Good

zone-8dd7

Broadcast latest alert (Telegram + IVR) sends a real Telegram message and places a real call — behind an are-you-sure confirmation and a server-side rate limit. Share cards for WhatsApp come from the Cell Story.

8.3 Cleanest air right now (step 3)

Tell residents what to do, in their language, on the channel they actually use — and let them tell you where it smells like smoke.

1 **Advisories by ward** 2 Send it 3 Clean-air routes 4 Citizen reports

1 Citizen Advisory English ?

App Telegram IVR call Big screen

how the same advisory reaches citizens on each channel

zone-00b3 moderate

Delhi zone-00b3: air is forecast moderate in +24h. Sensitive people (children, elderly, asthma) should limit long outdoor exertion.

zone-02c1 moderate

Delhi zone-02c1: air is forecast moderate in +24h. Sensitive people (children, elderly, asthma) should limit long outdoor exertion.

zone-02c3 moderate

Delhi zone-02c3: air is forecast moderate in +24h. Sensitive people (children, elderly, asthma) should limit long outdoor exertion.

zone-0e3d moderate

Delhi zone-0e3d: air is forecast moderate in +24h. Sensitive people (children, elderly, asthma) should limit long outdoor exertion.

2 Send it TELEGRAM · IVR · DISPLAY ?

 Broadcast latest alert (Telegram + IVR)

The same advisory, rendered per channel above; a WhatsApp-ready card for any place is one click away in its cell story.

3 Cleanest air right now LOWEST ~1 KM CELLS ?

48 Good

zone-8dd5

48 Good

zone-8dd7

The four lowest-PM2.5 ~1 km zones from the dense field, each with **Directions** ↗ (Google Maps) and a corridor exposure screen from the city centre — labelled a modelled guide, not a measurement.

8.4 Citizen reports (step 4)

Tell residents what to do, in their language, on the channel they actually use — and let them tell you where it smells like smoke.

1 Advisories by ward 2 Send it 3 Clean-air routes 4 Citizen reports

1 Citizen Advisory English ?

App Telegram IVR call Big screen

how the same advisory reaches citizens on each channel

zone-00b3 moderate

Delhi zone-00b3: air is forecast moderate in +24h. Sensitive people (children, elderly, asthma) should limit long outdoor exertion.

zone-02c1 moderate

Delhi zone-02c1: air is forecast moderate in +24h. Sensitive people (children, elderly, asthma) should limit long outdoor exertion.

zone-02c3 moderate

Delhi zone-02c3: air is forecast moderate in +24h. Sensitive people (children, elderly, asthma) should limit long outdoor exertion.

zone-0e3d moderate

Delhi zone-0e3d: air is forecast moderate in +24h. Sensitive people (children, elderly, asthma) should limit long outdoor exertion.

2 Send it TELEGRAM · IVR · DISPLAY ?

 Broadcast latest alert (Telegram + IVR)

The same advisory, rendered per channel above; a WhatsApp-ready card for any place is one click away in its cell story.

3 Cleanest air right now LOWEST ~1 KM CELLS ?

48 Good

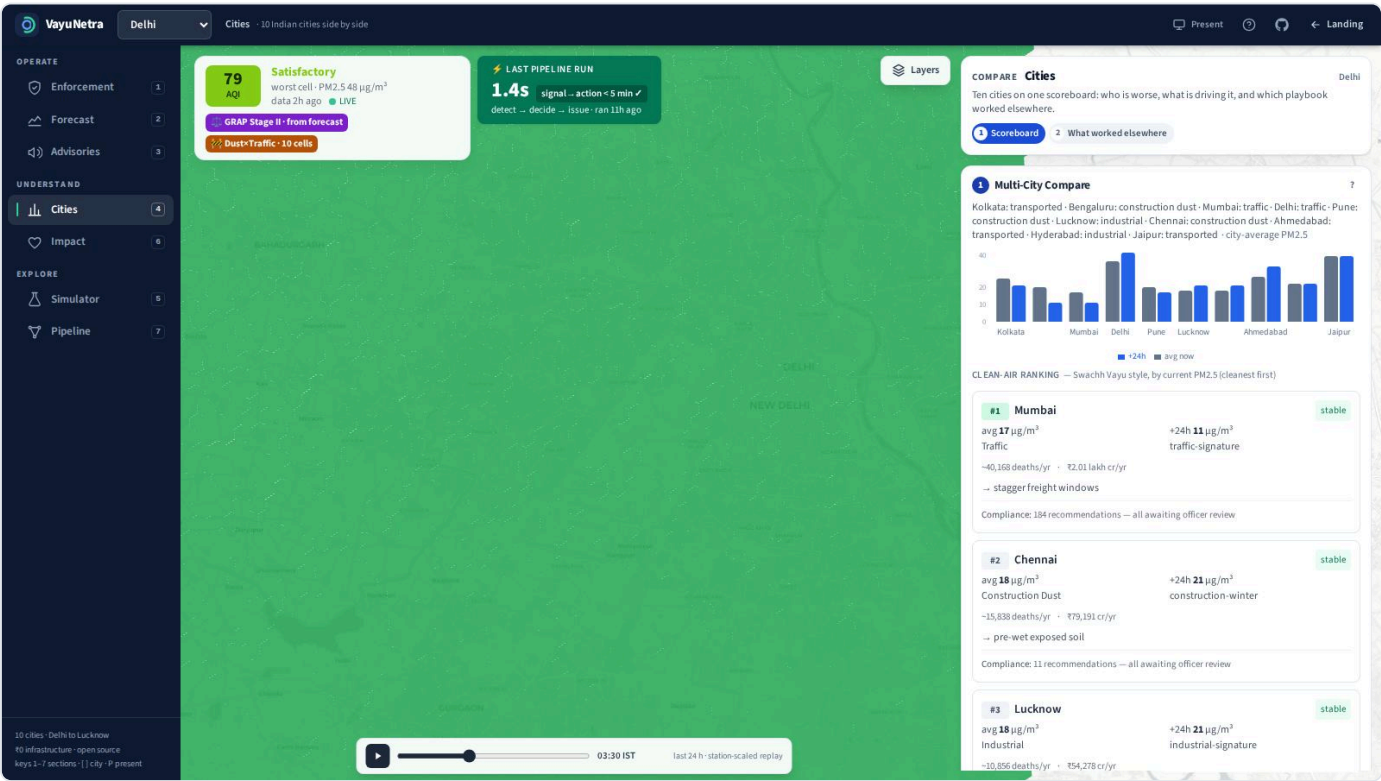
zone-8dd5

48 Good

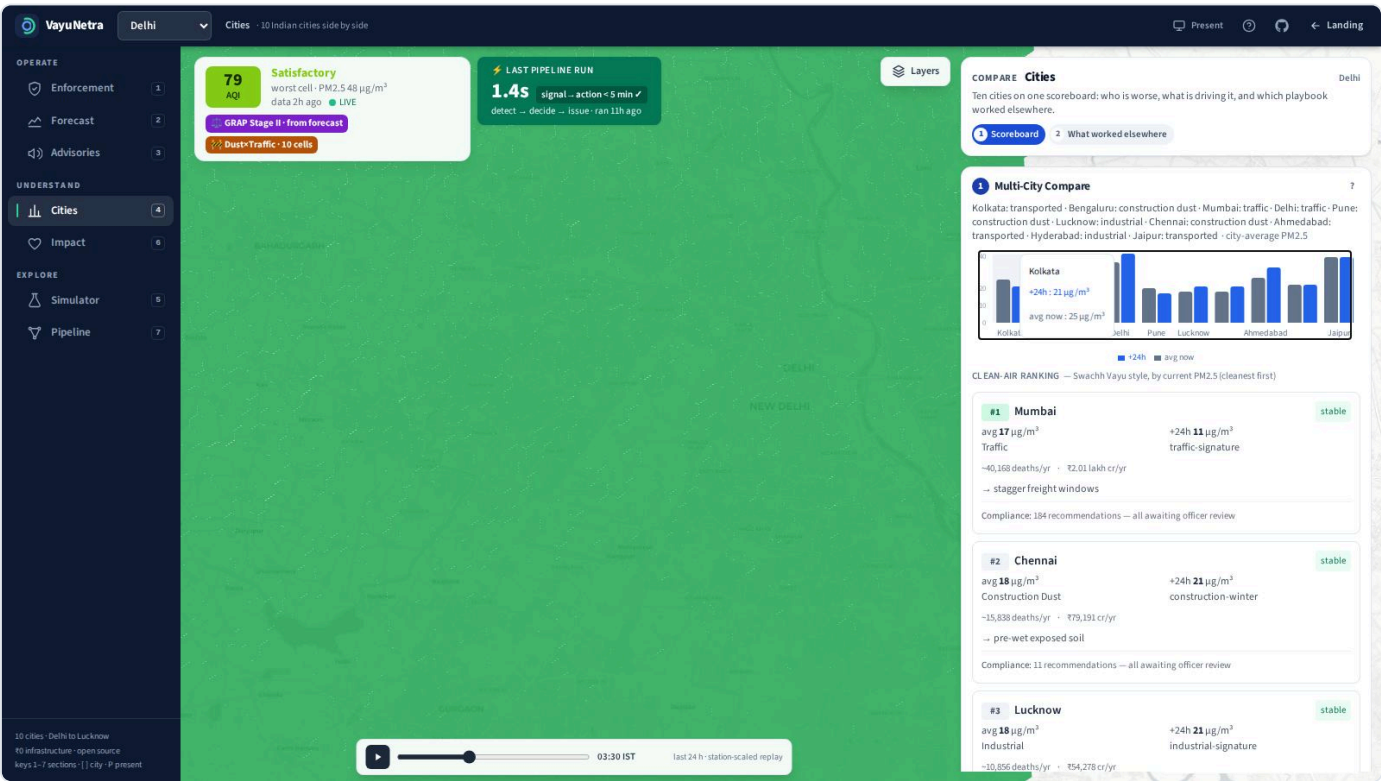
zone-8dd7

Report a pollution source: photo + category + location → a public list with a 72-hour SLA clock (received → verified → actioned → resolved / rejected; breaches in red). An officer marking a report *verified* registers the location as a candidate source so the next enforcement run scores it — the loop closes from the citizen side.

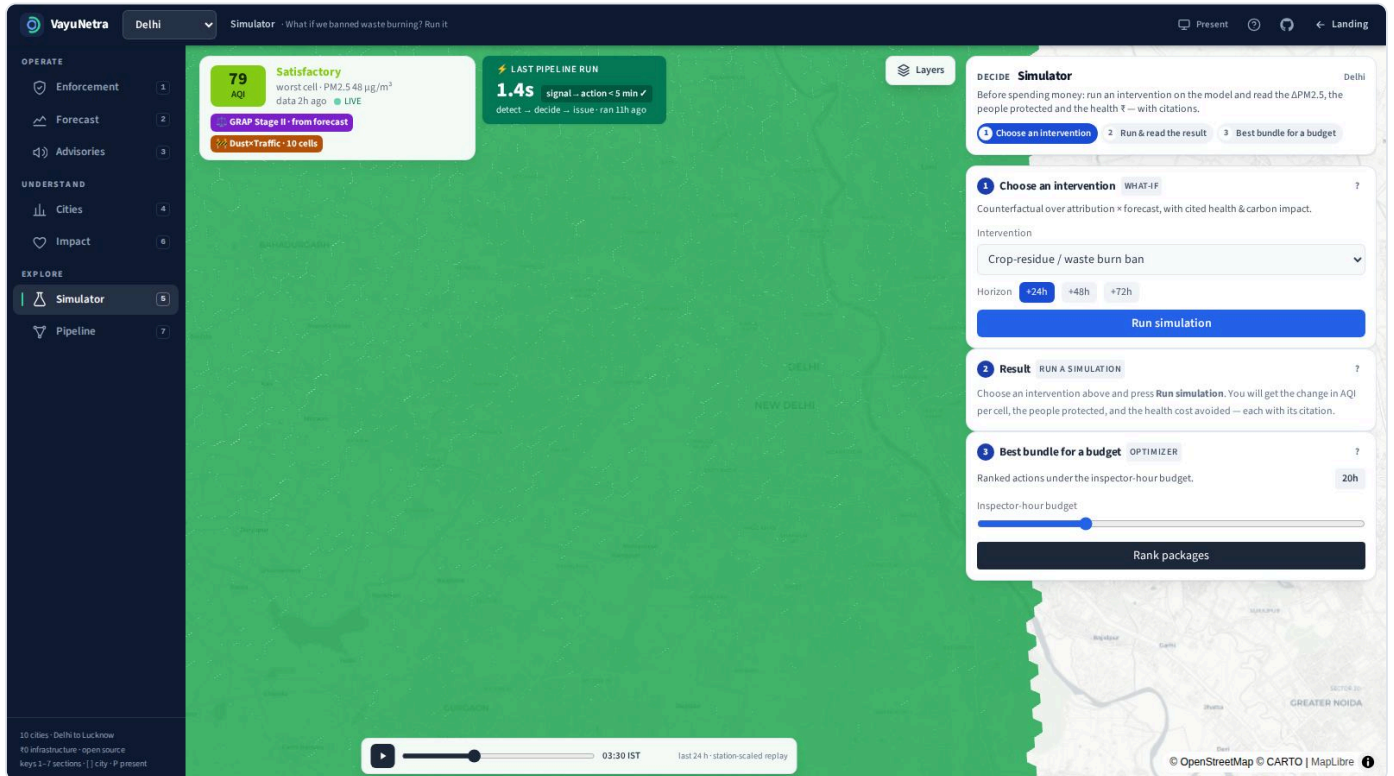
9. Cities (section 4)



Ten cities on one scoreboard: PM2.5 now vs +24 h, Swachh-Vayu-style ranking, trend, dominant source, playbook recommendation, health-burden line and enforcement-compliance counts — powered by the Multi-City agent. **Click a city** and the whole console follows:



10. Simulator (section 5)



Steps: **1 Choose an intervention** · **2 Run & read the result** · **3 Best bundle for a budget**. Pick an intervention (waste-burn ban · halt construction dust · odd-even traffic · industrial shutdown · GRAP Stage III package) and a horizon → **Run simulation**.

Before spending money: run an intervention on the model and read the $\Delta\text{PM}_{2.5}$, the people protected and the health ₹ — with citations.

1 Choose an intervention 2 Run & read the result 3 Best bundle for a budget

1 Choose an intervention WHAT-IF ?

Counterfactual over attribution $\times$ forecast, with cited health & carbon impact.

Intervention

Crop-residue / waste burn ban

Horizon +24h +48h +72h

Run simulation

2 Result +24H · 23 CELLS ?

Near-zero effect — honest result: this source contributes ~0% of the city's current $\text{PM}_{2.5}$ mix (see the blame map), so banning it changes little today. Try an intervention matching the dominant source, e.g. a traffic or construction measure.

avg ΔAQI 0 best cell ΔAQI 0
cells affected 23 confidence 40%

PEOPLE PROTECTED 0	HEALTH COST AVOIDED ₹0
DEATHS AVERTED 0.00 premature, over the window	CO ₂ E CO-BENEFIT 1,242 t vs cited emission factor

[sources \(5\) — every figure is cited](#)

3 Best bundle for a budget OPTIMIZER ?

Ranked actions under the inspector-hour budget. 20h

Inspector-hour budget

Rank packages

The result: ΔAQI (average and best cell), cells affected, confidence, and the cited impact cards — people protected (real population), health cost avoided (₹), deaths averted, CO₂e co-benefit — every figure with its citation. **Honest zero:** if the chosen source barely contributes today the app says so instead of inventing an impact.

Before spending money: run an intervention on the model and read the Δ PM2.5, the people protected and the health ₹ — with citations.

1 Choose an intervention 2 Run & read the result 3 Best bundle for a budget

1 Choose an intervention WHAT-IF ?

Counterfactual over attribution $\times$ forecast, with cited health & carbon impact.

Intervention

Crop-residue / waste burn ban

Horizon +24h +48h +72h

Run simulation

2 Result +24H · 23 CELLS ?

Near-zero effect — honest result: this source contributes ~0% of the city's current PM2.5 mix (see the blame map), so banning it changes little today. Try an intervention matching the dominant source, e.g. a traffic or construction measure.

avg Δ AQI 0 best cell Δ AQI 0
cells affected 23 confidence 40%

PEOPLE PROTECTED 0 HEALTH COST AVOIDED ₹0

DEATHS AVERTED 0.00 CO₂E CO-BENEFIT 1,242 t
premature, over the window vs cited emission factor

[sources \(5\) — every figure is cited](#)

3 Best bundle for a budget OPTIMIZER ?

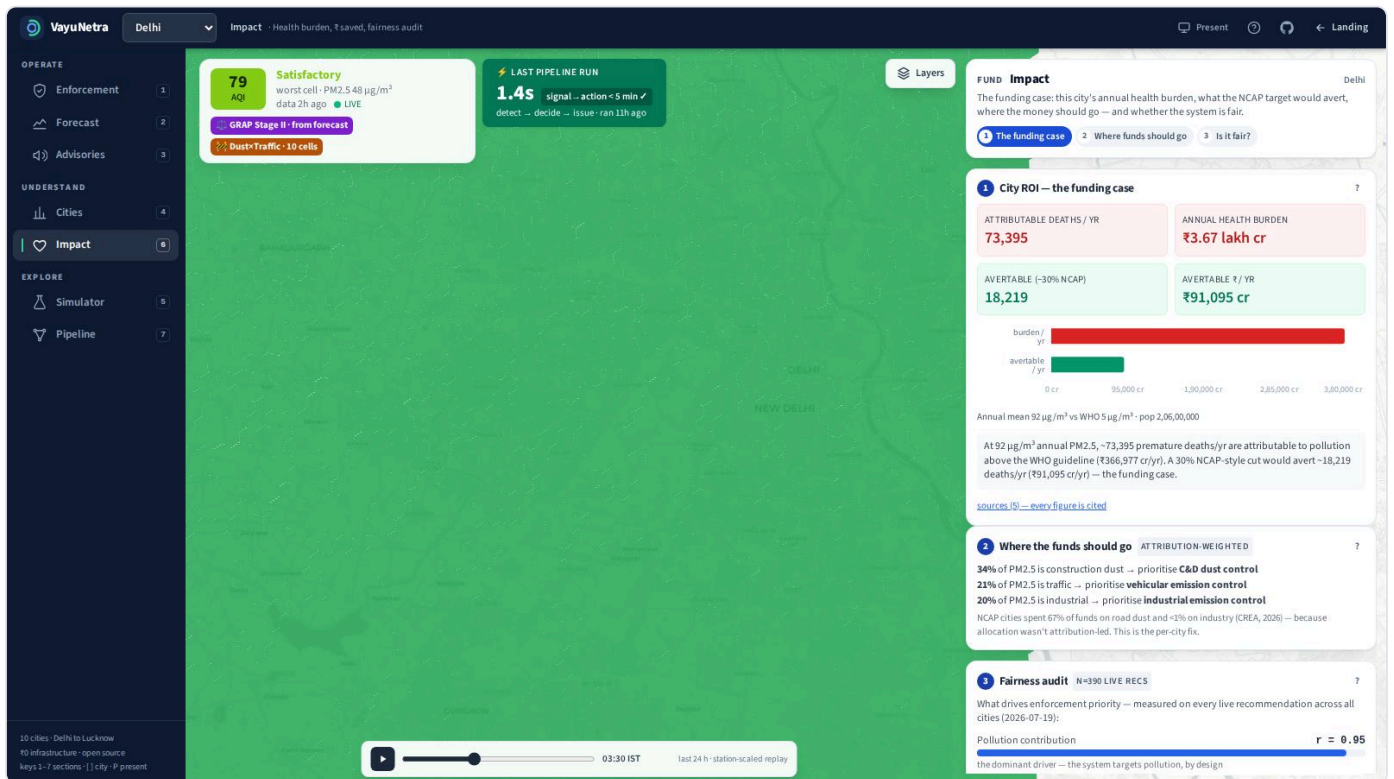
Ranked actions under the inspector-hour budget. 20h

Inspector-hour budget

Rank packages

Best bundle for a budget: set an inspector-hour budget → **Rank packages** → intervention bundles ranked by impact per inspector-hour.

11. Impact (section 6)



Steps: **1 City ROI** • **2 Where the funds should go** • **3 Fairness audit**. - **City ROI** — attributable deaths / yr, annual health burden (₹), what a 30 % NCAP cut would avert, with citations and the VSL caveat.

The funding case: this city's annual health burden, what the NCAP target would avert, where the money should go — and whether the system is fair.

- 1 The funding case
- 2 Where funds should go
- 3 Is it fair?

1 City ROI — the funding case

ATTRIBUTABLE DEATHS / YR

73,395

ANNUAL HEALTH BURDEN

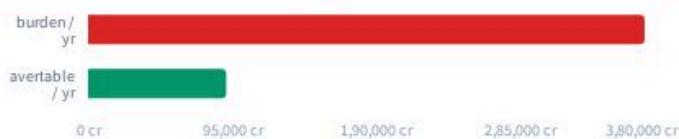
₹3.67 lakh cr

AVERTABLE (~30% NCAP)

18,219

AVERTABLE ₹ / YR

₹91,095 cr



Annual mean $92 \mu\text{g}/\text{m}^3$ vs WHO $5 \mu\text{g}/\text{m}^3$ · pop 2,06,00,000

At $92 \mu\text{g}/\text{m}^3$ annual PM_{2.5}, ~73,395 premature deaths/yr are attributable to pollution above the WHO guideline (₹366,977 cr/yr). A 30% NCAP-style cut would avert ~18,219 deaths/yr (₹91,095 cr/yr) — the funding case.

[sources \(5\)](#) — every figure is cited

2 Where the funds should go

ATTRIBUTION-WEIGHTED

34% of PM_{2.5} is construction dust → prioritise **C&D dust control**

21% of PM_{2.5} is traffic → prioritise **vehicular emission control**

20% of PM_{2.5} is industrial → prioritise **industrial emission control**

NCAP cities spent 67% of funds on road dust and <1% on industry (CREA, 2026) — because allocation wasn't attribution-led. This is the per-city fix.

3 Fairness audit

N=390 LIVE RECS

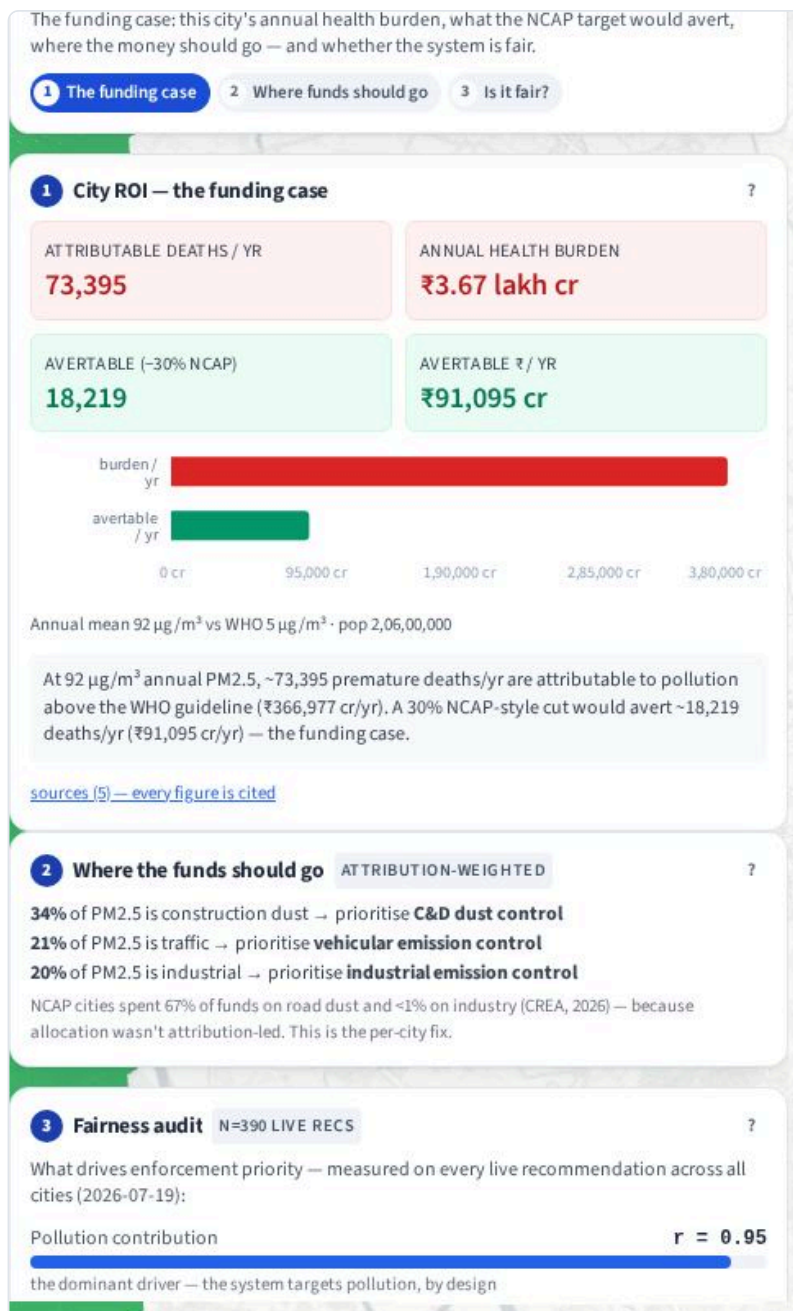
What drives enforcement priority — measured on every live recommendation across all cities (2026-07-19):

Pollution contribution

$r = 0.95$

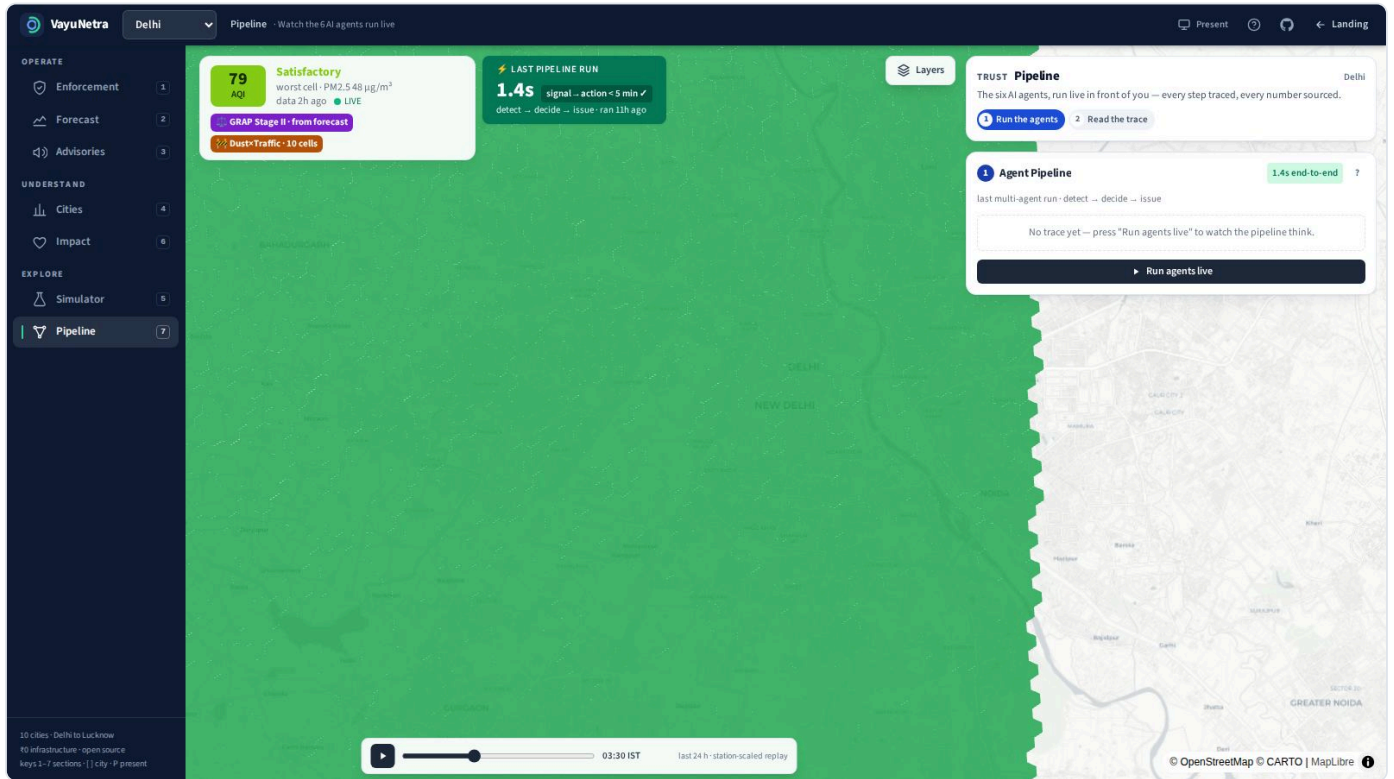
the dominant driver — the system targets pollution, by design

- **Where the funds should go** — live source shares mapped to NCAP spending heads (attribution-led allocation, the answer to CREA's finding that cities put 67 % of funds into road dust).

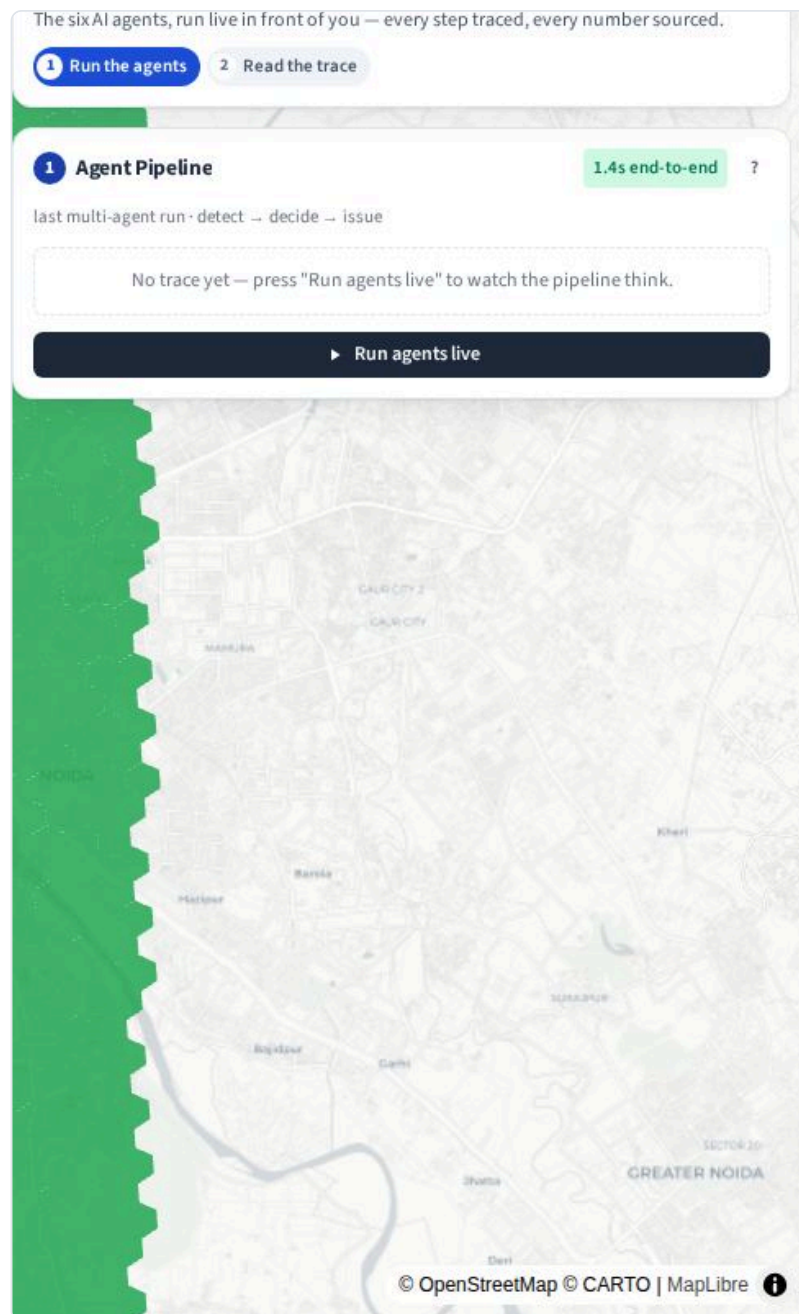


- **Fairness audit** — enforcement priority correlates with pollution contribution (by design) and people exposed (disclosed); **no socio-economic input exists anywhere in the pipeline.**

12. Pipeline (section 7)



Steps: **1 Run the agents** • **2 Read the trace**. The last multi-agent run as a timeline — Orchestrator → Attribution → Forecast → **Spike gate** → Enforcement → Advisory — with per-node durations. The spike gate is a decision: no spike → straight to advisory; spike → escalated to enforcement (on clean air Enforcement shows *skipped* rather than pretending). **Run agents live** triggers a real end-to-end run against the live database and updates the trace in front of you (0.8–9.7 s of compute).



13. Under the hood

13.1 The six agents (LangGraph)

Agent	Does
A0 Orchestrator	Reads latest signals, finds spiking cells, routes the graph, stamps per-node latency traces
A1 Attribution	GBM + SHAP hybrid source attribution per cell; abstains to signature priors below the R^2 gate
A2 Forecast	24/48/72 h per-cell PM2.5, persistence-blended, CQR-calibrated intervals, calibrated exceedance probabilities
Spike gate	Branches: enforcement only when cells spike ($PM_{2.5} > 120 \mu g/m^3$ or focus cells exist)
A3 Enforcement	Priority = contribution $\times$ exposure $\times$ actionability $\times$ confidence; jurisdiction-aware RAG citations; writes the worklist without touching acted-upon items
A4 Advisory	Deterministic multilingual advisories, vulnerability-adjusted per ward/audience

Agent	Does
A5 Multi-City	Cross-city trends, dominant sources, playbooks, health burden

Supporting models: E1 satellite source detection (labelled Earth-Engine heuristic; a learned detector is roadmap) · E2 dense 1-km field · E3 counterfactual what-if · E5 optimiser · E7 health ₹ / lives / CO₂e · GRAP forecast trigger · multimodal RAG (text + Sentinel-2 patches).

13.2 Data sources (all free / open)

CPCB CAAQMS (via OpenAQ / data.gov.in) · Open-Meteo + ERA5 weather and boundary layer · Sentinel-5P NO₂, Sentinel-2, NASA FIRMS (Earth Engine) · OpenStreetMap sources / facilities / roads · GPW v4.11 population · GTFS + road-network mobility proxy. Ingest and model refresh run on GitHub Actions (hourly + nightly); one row per reading (unique key), raw readings kept 180 days then archived to Storage with a daily rollup kept forever (docs/SCALE.md).

13.3 API (FastAPI on Render)

44 HTTP routes + 1 WebSocket, one envelope. Feeds (/cities /aqi/current /history* /attribution /forecast /coverage /plume /static-layers ...), officer (/enforcement*, /interventions*, /brief*), citizen (/advisory*, /clean-zones, /telegram/webhook, /ivr/*, /report*), analysis (/comparison /roi /simulate /optimize /latency /traces), metrics (/metrics/benchmark /metrics/attribution /metrics/interventions), /landing/snapshot, agents/admin (/agent/query, /admin/cities, WS /live). Full table with roles: docs/API_CONTRACT.md . Public reads via anon JWT under RLS; every write server-side with the service role; admin behind X-Admin-Key .

13.4 Database (Supabase Postgres + PostGIS + pgvector)

cities, measurements, attribution, forecasts, emission_sources, enforcement_recgs, enforcement_status_log, intervention_tracking, advisories, advisory_subscribers, citizen_reports, kb_chunks, action_traces, coverage_field, vulnerability, profiles, pm25_daily_rollup — 20 migrations in supabase/migrations/ .

13.5 Honesty, by construction

- Attribution **abstains** (R² gate + coverage mask) instead of guessing; how today's split was made is shown in the console.
- The forecast is benchmarked the way a reviewer would (strict temporal split, production-faithful rolling refit, three baselines, shared support mask, regime slices, onset recall, Brier skill) and the numbers — including the Severe-tail weakness — are served by the API and printed in the console.
- Attribution is compared bucket by bucket with published apportionment (TERI-ARAI 2018; Guttikunda 2019, CSTEP 2022) with disagreements stated (docs/ATTRIBUTION_VALIDATION.md).
- Real GRAP orders are replayed and a weather-normalised effect check is published, null results included (docs/OUTCOMES.md).
- Health text is LLM-free; impact figures return null over invented constants; notices are drafts; nothing fabricated is written to the production database; the audit trail is append-only.
- Where ₹0 stops is measured and published (docs/SCALE.md).

13.6 Ops

make dev · make test (204 backend tests, 64 % line coverage, CI gate 55 %) · cd web && npx playwright test (7 smoke; VN_LIVE=1 adds the 9-flow live officer journey) · web/scripts/qa/*.mjs (full walkthrough, axe, mobile, city sweep, deck render) · make prewarm (GO / NO-GO) · make benchmark · GitHub Actions: hourly ingest, nightly models / registry / enforcement / advisories / brief / retention.

14. Troubleshooting & FAQ

Symptom	Meaning / fix
Amber "backend waking up" notice	Render free-tier cold start; the console shows bundled fixtures meanwhile. retry or wait — it clears on the first successful call; the notice never blocks the map controls. make prewarm before demos.
Map loads, panels empty	Backend still waking, or a city with thin data — switch city once.
"No per-cell forecast" in the Cell Story	That cell has no fresh per-cell rows; the city Forecast section still works.
Worklist says nothing to enforce	Air is clean (spike gate) or filters exclude everything.
Advisory empty in a language	That city does not publish that language (languages come from the city config).
Notice PDF takes seconds	Live dossier assembly (RAG + satellite patch) — expected.
Tour will not reappear	Once-only by design; replay with ? .
Simulator says "near-zero effect"	Honest result — pick the dominant source.
Real or demo?	<code>/health</code> shows <code>DEMO_MODE</code> ; the landing page and console say "as of" and "snapshot" where a fixture is shown.

Glossary. **H3** — Uber's hexagonal grid; res-8 $\approx 1 \text{ km}^2$. **SHAP** — per-feature contribution of a prediction, in $\mu\text{g}/\text{m}^3$. **CQR** — conformalised quantile regression; makes the 80 % band cover ~ 80 %. **P(>120)** — calibrated probability the cell exceeds Very Poor. **Persistence** — the "tomorrow = today" baseline. **Onset recall** — share of clean→Very-Poor transitions the alarm flags in advance. **GRAP** — Delhi-NCR's Graded Response Action Plan (Stages I–IV). **RAG** — retrieval over the regulation corpus (citations, not free text). **RLS** — Postgres row-level security. **PRANA** — the NCAP reporting portal cities report actions and spend to. **GPW** — Gridded Population of the World v4.11.

15. Repo map

Path	What
<code>api/main.py</code>	All FastAPI routes
<code>agents/</code>	The six LangGraph agents, the brief, the notice PDF writer
<code>ml/</code> , <code>core/</code>	Models, benchmark and hindsight harnesses, H3 utils, impact and intervention math, city configs, ward lookup
<code>connectors/</code>	CPCB / OpenAQ, Earth Engine, Open-Meteo, OSM, population, vulnerability, mobility
<code>scripts/</code>	prewarm, refresh_advisories, archive_measurements, morning brief, fetch_history, doc PDFs
<code>web/src/</code>	React console + landing (<code>App.tsx</code> shell, one panel per card, <code>console/flows.tsx</code> = the spines)
<code>web/scripts/qa/</code>	Playwright QA: full walkthrough (this guide's screenshots), axe, mobile, city sweep, deck render
<code>demo/fixtures/</code>	The 19 JSON fixtures behind DEMO_MODE
<code>supabase/migrations/</code>	Schema, RLS, seeds, closure log, one-row-per-reading key, retention rollup (20 migrations)

Path	What
<code>docs/</code>	This guide (<code>guide/</code> = its screenshots), the finale deck (<code>VayuNetra_Pitch.html</code> + static exports, <code>PITCH_SCRIPT.md</code> , built from <code>docs/pitch/</code>), benchmarks and outcomes, API contract, validation, scale, QA evidence
<code>eval/evaluate.ipynb</code>	The validation trail, including the documented failures